

INSTANTON AND PILLOWCASE HOMOLOGY OF THE $(-2, 3, q)$ PRETZEL KNOTS

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ABSTRACT. Hedden, Herald and Kirk conjectured that every knot admits a decomposition along a Conway sphere for which the Lagrangian Floer homology of the two associated immersed curves in the pillowcase recovers Kronheimer and Mrowka's reduced singular instanton homology. We prove the conjecture for the hyperbolic pretzel knots $P(-2, 3, q)$; it was previously known for two-bridge knots and for some torus knots. The decomposition is obtained from Hedden, Herald and Kirk's decomposition of the torus knot $T(3, 5)$ by Dehn twists along the Conway sphere, which act on the pillowcase by a linear shear. Building on our analysis of the torus knots $T(3, n)$, we compute the Floer complex with zero bounding cochain and its $\mathbb{Z}/4$ grading by hand; its differential is nonzero for every member of the family except $P(-2, 3, 7)$. The same method proves it for a family of twisted torus knots, under a hypothesis on the chirality of the torus-knot decomposition. On the instanton side, we show that the instanton homology of $P(-2, 3, q)$ is free abelian of rank $q + 2$ for every odd $q \geq 3$, extending Lobb and Zentner's rational computation to the integers by means of Manion's integral Khovanov homology; for the hyperbolic members this also follows from work of Daemi and Scaduto.

1. INTRODUCTION

1.1. Two homologies for one knot. Let $K \subset S^3$ be a knot. On the gauge-theoretic side sits the reduced singular instanton knot homology $I^\natural(K)$ of Kronheimer and Mrowka [KM11a, KM11b]. Its rank is bounded below by the coefficient norm of the Alexander polynomial and above by the rank of reduced Khovanov homology (3.1). On the symplectic side sits the pillowcase program of Hedden, Herald and Kirk [HHK14, HHK18]. Write $K = (Y, T) \cup (D, U)$ as the union of a tangle and the trivial tangle along a Conway sphere. The traceless $SU(2)$ character variety of the four-punctured sphere is the *pillowcase*, a two-sphere with four orbifold points, and after a holonomy perturbation each side determines an immersed curve in its smooth part: the tangle side a curve L_1 , and the trivial side, with an earring, the curve L_0 , which depends on a small earring parameter $\epsilon > 0$ (§2.2). Their Lagrangian Floer homology $H^\natural(Y, T, \pi)$ is defined when L_1 is restricted and the pair is admissible (Definitions 2.1 and 2.2), and Hedden, Herald and Kirk conjecture that for suitable perturbations it is isomorphic to $I^\natural(K)$ [HHK18, Conj. 6.5] (Conjecture 2.5), and they compare the $\mathbb{Z}/4$ gradings of the two sides [HHK18, §§6.1, 11]. For general decompositions figure-eight bubbles at the self-intersections of the curves deform the differential. Cazassus, Herald, Kirk and Kotelskiy conjecture that each tangle is assigned an immersed curve together with a bounding cochain, and that the deformed theory recovers $I^\natural(K)$ [CHKK22, Conj. D]; conditional on this conjecture, Smith proves that the cochain on one Conway-sum tangle in $P(-2, 3, 5)$ cannot vanish [Smi24, proof of Thm. 5.9].

This paper studies both sides for the pretzel knots $K_q = P(-2, 3, q)$, $q \geq 3$ odd (§2.5). The smallest members are the torus knots $K_3 = T(3, 4)$ and $K_5 = T(3, 5)$, whose instanton and pillowcase homology are treated in [Wue26a, Wue26b, Wue26c], and for $q \geq 7$ the knot K_q is

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hyperbolic. We decompose K_q by regluing the tangle of Hedden, Herald and Kirk for $T(3, 5)$ to the trivial tangle by Dehn twists along the Conway sphere. For this decomposition we compute the pillowcase homology with zero bounding cochain and find that it agrees with $I^{\natural}(K_q)$; this proves the conjecture of Hedden, Herald and Kirk for every K_q , and the same method applies to a family of twisted torus knots (Theorems 1.2 and 1.3). Smith decomposes the same knots as a sum of rational tangles, with the earring on the -2 band. For that decomposition the pairing with zero bounding cochain has the wrong rank at $q = 5$ [Smi24, proof of Thm. 5.9], and also at $q = 7$ for a piecewise-linear model of Smith's curve [Wue26d]. The companion paper [Wue26d] constructs, for $q = 5, 7$, Maurer–Cartan elements that correct the rank, in the two-arc algebra of Kotelskiy, Watson and Zibrowius [KWZ19]; the instanton dimensions of Theorem 1.1 are its targets.

1.2. The instanton side.

Theorem 1.1. *For every odd $q \geq 3$, $I^{\natural}(K_q; \mathbb{Z}) \cong \mathbb{Z}^{q+2}$. For $q \geq 7$ the signature is $\sigma(K_q) = -(q+1)$, and in Kronheimer and Mrowka's absolute grading the graded ranks of $I^{\natural}(K_q)$ are*

$$(1 + N_3, N_1, N_1, N_3) \quad \text{in degrees } \text{gr } -\sigma(K_q) = 0, 1, 2, 3, \quad N_1 = \lceil \frac{q+1}{4} \rceil, \quad N_3 = \lfloor \frac{q+1}{4} \rfloor.$$

The rank over $\mathbb{Q}$ is due to Lobb and Zentner [LZ20, Prop. 4.7]. For $q \geq 7$ the freeness and the graded ranks follow from the computation of the irreducible instanton homology of K_q by Daemi and Scaduto [DS24b, Prop. 8.30] together with their integral spectral sequence (Proposition 3.6). Our proof of freeness is uniform in q : it runs Lobb and Zentner's two-sided rank bound with integer coefficients, using Manion's integral computation of Khovanov homology [Man18] and the integral spectral sequence of Kronheimer and Mrowka. For $q = 3, 5$ the graded ranks are those of the torus knots in [Wue26c]. Section 3 gives the proof and Remark 3.4 the remaining attributions.

1.3. The conjecture of Hedden, Herald and Kirk for the pretzel knots. Hedden, Herald and Kirk compute the pillowcase homology of $T(3, 5)$ from a tangle (Y, T) whose complement has free fundamental group of rank two, glued to the trivial tangle [HHK18, §§10–11]. Let t_e be the Dehn twist along the curve of the Conway sphere that separates two of the punctures from the other two. Regluing by t_e^{-m} adds m full twists on two strands, and the resulting knot is K_{5+2m} (Theorem 4.4). That the original gluing gives the positive torus knot $T(3, 5)$ is verified by a computer-assisted argument (Remark 4.3). The unperturbed character variety $R(Y, T)$ is the union of an arc and a circle that meet in two points $c_{\pm}$ (§4.3). The holonomy perturbations on the tangle side are those of [HHK18, §10] used in [Wue26c]: they are supported near two curves in Y , with perturbation functions $\varepsilon_A \sin$ and $\varepsilon_B \sin$, and items (P1)–(P9) in §§4.3–4.4 recall the facts about them that we use from [Wue26c]. For $a > 0$ the closed cone $\mathcal{C}_a$ is the set of parameters $\varepsilon = (\varepsilon_A, \varepsilon_B)$ such that, at both $c = c_+$ and $c = c_-$, the linear part of $\kappa_c(\varepsilon)$, the critical value near c of the function to which the equations of the perturbed variety reduce near c , has absolute value at least $a|\varepsilon|$ [Wue26c, Def. 5.5]; it is the complement of an open conical neighborhood of the two lines $\{\varepsilon_B = \mp\sqrt{3}\varepsilon_A\}$ (Proposition 4.10). The generator r_+ below is the intersection point of L_0 and L_1 near the corner $(0, 0)$ that comes from the abelian representation of the knot group (§2.4).

Theorem 1.2. *Let $q \geq 7$ be odd, and let $K_q = (Y, T) \cup_{t_e^{-(q-5)/2}} (D, U)$ be the decomposition above. Fix $a > 0$. For every sufficiently small $\varepsilon \in \mathcal{C}_a \setminus \{0\}$ and every sufficiently small earring parameter:*

- (1) L_1 is a restricted immersed 1-manifold, and (L_0, L_1) is admissible;
- (2) the Floer complex has $q + 2 + 4N_q$ generators, where $N_q = \lfloor 5(q-6)/12 \rfloor - \lfloor (q-6)/12 \rfloor$;

- (3) *the differential has rank $2N_q$, and it is nonzero for $q \geq 9$;*
- (4) *with the absolute grading normalized by $\text{gr}(r_+) = \sigma(K_q)$, as in [HHK18, (19)], and with any absolute grading on the circle component of L_1 when there is one, there is an isomorphism $H^\natural(Y, T, \pi) \cong I^\natural(K_q; \mathbb{F}_2)$ of $\mathbb{Z}/4$ -graded vector spaces.*

In particular, the conjecture of Hedden, Herald and Kirk [HHK18, Conj. 6.5] holds for every K_q .

The perturbation resolves the two double points $c_\pm$ of $R(Y, T)$ in one of two ways, joined for $|\varepsilon_B| < \sqrt{3}|\varepsilon_A|$ and split for $|\varepsilon_B| > \sqrt{3}|\varepsilon_A|$ (§4.4 and Proposition 4.10); for small a both occur in $\mathcal{C}_a$, and the theorem holds in both. For $q \geq 9$ the complex is not minimal. Hedden, Herald and Kirk verify the conclusion of their conjecture for two-bridge knots [HHK18, §7], a family that contains hyperbolic knots, and for several torus knots, among them $T(3, 4)$, $T(3, 5)$ and $T(3, 7)$ [HHK18, §§10–11]; the torus knots $T(3, n)$ are treated in [Wue26c]. Fukumoto, Kirk and Pinzón-Caicedo describe the unperturbed character variety of the pretzel knots for a decomposition of their own and count 9 generators for $P(-2, 3, 7)$ [FKP17, §5] (Remark 4.19). For $q \geq 7$ the knot K_q is hyperbolic and not two-bridge (Proposition 4.5). To our knowledge, these are the first hyperbolic knots outside the two-bridge family for which the conjecture is verified, with the $\mathbb{Z}/4$ gradings and nonzero Floer differentials.

The same regluing applies to the tangles of [HHK18, §10] for $T(3, n)$. The *twisted torus knot* $T(3, n; 2, m)$ is the closure of the braid $(\sigma_1 \sigma_2)^n \sigma_1^{2m}$.

Theorem 1.3. *Let $n \geq 7$ with $n \equiv 1$ or $2 \pmod{6}$, let $m \geq 1$, and let $K = T(3, n; 2, m)$. Then $\sigma(K) = \sigma(T(3, n)) - 2m$, and $I^\natural(K; \mathbb{Z})$ is free of rank $|\sigma(K)| + 1$. Assume, as in [HHK18] and [Wue26c], that the tangle of [HHK18, §10] for $T(3, n)$ glued to the trivial tangle is the positive torus knot $T(3, n)$. Then this tangle, reglued by t_e^{-m} , is a decomposition of K , and for small holonomy perturbations and generic small earring parameters it satisfies the conclusion of [HHK18, Conj. 6.5]: $H^\natural \cong I^\natural(K; \mathbb{F}_2)$ as $\mathbb{Z}/4$ -graded vector spaces.*

Theorem 5.1 gives the precise form, with the graded ranks. The family of Theorem 1.3 contains torus knots, $T(3, 6j + 1; 2, 1) = T(3, 6j + 2)$, and by braid conjugacy $T(3, 6j + 2; 2, m) = T(3, 6j + 1; 2, m + 1)$ (Lemma 5.3). The other members, $T(3, 6j + 1; 2, M)$ with $j \geq 1$ and $M \geq 2$, are not torus knots, not alternating and not pretzel knots $P(-2, 3, q)$ (Lemma 5.6). We make no claim about their hyperbolicity.

1.4. Outline of the arguments. *Theorem 1.2.* The regluing replaces L_1 by its image under the linear shear $(\gamma, \theta) \mapsto (\gamma, \theta + (q - 5)\gamma)$ of the pillowcase (Lemma 4.6). The unperturbed character variety of the $T(3, 5)$ tangle is the union of an arc of binary dihedral characters and a circle of characters, the *central circle*, meeting it in two points. A word w in the tangle group with $\rho(w) = 1$ for every representation ρ on the central circle shows that its image in the pillowcase is a union of straight segments (Lemma 4.7 and Corollary 4.8). The perturbed curves are then those of [Wue26c], sheared, and on them we compute the intersection points with L_0 , their $\mathbb{Z}/4$ gradings, and the bigons by hand. The bigons come in chains along the lifts of the earring curve L_0 to $\mathbb{R}^2$, which gives the rank $2N_q$ of the differential (Lemma 4.16). The identification of the reglued knot uses only its fundamental group: the relator is that of Clay and Watson’s presentation of the twisted torus knots [CW13, Prop. 22] (Lemma 4.2), two-generator knots are prime [Nor82], knots are determined by their complements [GoLu89], and an induction on the signature fixes the chirality.

Theorem 1.3. For $n \equiv 1, 2 \pmod{6}$ the unperturbed variety is an arc with a straight image and $2\lfloor n/6 \rfloor$ circles that miss the edges of the pillowcase [Wue26c]. The sheared arc meets L_0 in $1 + 2M$ points and carries no bigon, and each circle contributes one dimension in each degree to the homology, by a theorem of Hedden, Herald and Kirk [HHK18, Thm. 5.4]. On the knot side,

a closed formula for the Burau matrix of the braid gives the signature and the coefficient norm of the Alexander polynomial, and the instanton rank follows from the work of Daemi and Scaduto.

1.5. Scope of claims. Theorem 1.1 is unconditional. Theorems 1.2 and 1.3 rest on the analysis of the perturbed curves of the torus-knot tangles in [Wue26c]. For Theorem 1.2 the base case is verified by computer (Remark 4.3); for the decomposition in Theorem 1.3 it is a hypothesis of the statement, shared with [HHK18] and [Wue26c], that the decomposition of $T(3, n)$ gives the positive torus knot, which the knot group cannot detect. Theorem 1.3 also uses the computation of Hedden, Herald and Kirk for circles that miss the edges [HHK18, Thm. 5.4], whose proof in [HHK18] rests on their invariance theorem [HHK18, Thm. 4.1] (Remark 2.4). Numerical traces of the perturbed curves agree with them (Remark 4.21) but are not used in the proofs.

1.6. Organization. Section 2 recalls the pillowcase Floer theory of Hedden, Herald and Kirk. Section 3 proves Theorem 1.1. Section 4 proves Theorem 1.2, and Section 5 proves Theorem 1.3. Section 6 discusses related work and open problems, and Appendix A describes the verified computation behind Remark 4.3 and the computations that corroborate the proofs.

Acknowledgements. The pillowcase model and the geometry used here are due to Hedden, Herald and Kirk [HHK14, HHK18] and to Fukumoto, Kirk and Pinzón-Caicedo [FKP17]. The closed-form Khovanov input is Manion's [Man18], the Alexander polynomials belong to Hironaka's family [Hir01], the presentation of the twisted torus knot groups is Clay and Watson's [CW13], the rational rank of the instanton homology is Lobb and Zentner's [LZ20], and the integral and graded instanton computations for $q \geq 7$ and for twisted torus knots also follow from the work of Daemi and Scaduto [DS24b].

2. BACKGROUND

This section fixes notation and collects the results from the literature that are used later. §§2.1–2.4 recall the Lagrangian Floer theory of Hedden, Herald and Kirk in the pillowcase [HHK18], which is the setting of Sections 4 and 5; results of [HHK18] are cited by their numbering in arXiv:1501.00028v1. §2.5 fixes the conventions for the knots. We use $SU(2)$ as the group of unit quaternions, so that a traceless element is a purely imaginary unit quaternion. Coefficients are in $\mathbb{F}_2$ unless stated otherwise.

2.1. The pillowcase. Let $G \cong \mathbb{Z}^2 \rtimes \mathbb{Z}/2$ be the group of isometries of $\mathbb{R}^2$ generated by $(\gamma, \theta) \mapsto (\gamma + 2\pi, \theta)$, $(\gamma, \theta) \mapsto (\gamma, \theta + 2\pi)$ and $(\gamma, \theta) \mapsto (-\gamma, -\theta)$. The *pillowcase* is the quotient $P = \mathbb{R}^2/G$, a topological 2-sphere. The quotient map $\mathbb{R}^2 \rightarrow P$ is a branched covering, branched over the images of the lattice points $(\pi\mathbb{Z})^2$, which are the four *corners* of P [HHK18, §3.1]. The rectangle $[0, \pi] \times [0, 2\pi]$ is a fundamental domain; we call $[0, \pi] \times [0, \pi]$ the front face, $[0, \pi] \times [\pi, 2\pi]$ the back face, and the images of the lines $\gamma \in \pi\mathbb{Z}$ the left and right edges. Put $P^* = (\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2)/G$, oriented by $d\gamma \wedge d\theta$. The restriction $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2 \rightarrow P^*$ is a regular covering with deck group G , so paths, loops and immersed discs in P^* lift to $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2$; we work with such lifts throughout. The double cover $\mathbb{R}^2/(2\pi\mathbb{Z})^2$ of P is a torus, and P is its quotient by the elliptic involution $(\gamma, \theta) \mapsto (-\gamma, -\theta)$.

The pillowcase is the traceless character variety of the four-punctured sphere. With the presentation $\pi_1(S^2 \setminus \{a, b, c, d\}) = \langle a, b, c, d \mid ba = cd \rangle$ of [HHK14, HHK18], the assignment

$$a \mapsto \mathbf{i}, \quad b \mapsto e^{\gamma\mathbf{k}}\mathbf{i}, \quad c \mapsto e^{\theta\mathbf{k}}\mathbf{i}, \quad d \mapsto e^{(\theta-\gamma)\mathbf{k}}\mathbf{i} \quad (2.1)$$

induces a homeomorphism of P with the space $R(S^2, \{a, b, c, d\})$ of conjugacy classes of $SU(2)$ representations sending each of a, b, c, d to a traceless element; the corners correspond to the abelian representations [HHK14, Prop. 3.1], [HHK18, Prop. 6.1]. We use these coordinates throughout.

On $\mathbb{R}^2$ we use the function

$$\varphi = \theta - \gamma.$$

Translations in G change φ by elements of $2\pi\mathbb{Z}$, and $(\gamma, \theta) \mapsto (-\gamma, -\theta)$ changes its sign. The *diagonal arc* $\Delta = \{(\gamma, \gamma) \mid \gamma \in [0, \pi]\} \subset P$ [HHK18, (12)] joins the corners $(0, 0)$ and (π, π) , and its preimage in $\mathbb{R}^2$ is the union of the lines

$$\mathcal{D}_k = \{\varphi = 2\pi k\}, \quad k \in \mathbb{Z}.$$

The lines $\{\varphi = \pi + 2\pi k\}$ are lifts of the arc of slope one on the back face, which is not part of Δ . The lattice points on $\mathcal{D}_k$ are the points $(j\pi, j\pi + 2\pi k)$, $j \in \mathbb{Z}$.

2.2. The earring curve. For $\epsilon > 0$ let $L_0 = L_0^{\epsilon, 0}: S^1 \rightarrow P^*$ be the immersion defined by

$$\tilde{L}_0(t) = (t + \epsilon \sin t + \frac{\pi}{2}, t - \epsilon \sin t + \frac{\pi}{2}), \quad t \in [0, 2\pi], \quad (2.2)$$

followed by the branched covering [HHK18, Def. 3.4, (13)]; as in [HHK18, §11] we take the perturbation function g of [HHK18, (13)] to be zero. It is the restriction to the boundary of the traceless character variety of the trivial tangle with an earring, after a small holonomy perturbation [HHK18, Thm. 6.2]. For small ϵ it is self-transverse with one double point, at $(\pi/2, \pi/2)$, it lies in an $O(\epsilon)$ -neighborhood of Δ , and it passes around the corners $(0, 0)$ and (π, π) through the back face.

Since $\tilde{L}_0(t + 2\pi) = \tilde{L}_0(t) + (2\pi, 2\pi)$, the preimage of L_0 in $\mathbb{R}^2$ is the union of the curves

$$S_+^k = \tilde{L}_0 + (0, 2\pi k), \quad S_-^k = -\tilde{L}_0 + (0, 2\pi k), \quad k \in \mathbb{Z},$$

which we call the *lifts* of L_0 . Each is a properly embedded line within $\sqrt{2}\epsilon$ of $\mathcal{D}_k$, and a translation by $(2\pi a, 2\pi b)$ carries $S_\pm^k$ to $S_\pm^{k+b-a}$. Reparametrizing, $S_-^0 = \{(t + \frac{\pi}{2} - \epsilon \sin t, t + \frac{\pi}{2} + \epsilon \sin t)\}$. The slope of S_+^k at the point with parameter t is $(1 - \epsilon \cos t)/(1 + \epsilon \cos t)$, and that of S_-^k is its reciprocal. Hence on the front strip $0 < \gamma < \pi$ the lift S_+^k has slope less than 1, and S_-^k has slope greater than 1 for $\epsilon < \gamma < \pi - \epsilon$ (it has slope 1 at $\gamma = \epsilon$ and $\gamma = \pi - \epsilon$); in P these are the two branches of L_0 along the interior of Δ [HHK18, §5.1]. The two lifts near $\mathcal{D}_k$ cross exactly at the points $(\frac{\pi}{2} + j\pi, \frac{\pi}{2} + j\pi + 2\pi k)$, and they pass each lattice point of $\mathcal{D}_k$ on opposite sides at distance $\sqrt{2}\epsilon$: for instance S_+^0 passes $(0, 0)$ through $(-\epsilon, \epsilon)$ and S_-^0 through $(\epsilon, -\epsilon)$, and the sides alternate from one lattice point of $\mathcal{D}_k$ to the next.

2.3. Restricted curves and the Floer complex. Let ℓ_1 be the line field of slope one on $\mathbb{R}^2$; it descends to P^* . For a path α immersed in P^* , the Maslov index $\mu(\alpha, \ell_1)$ is the signed number of times the tangent line of α passes ℓ_1 , counterclockwise passages counting +1, with the half-open endpoint convention of [HHK18, §2.3]. For lines ℓ_0, ℓ'_1, ℓ , pairwise transverse, the *triple index* $\tau(\ell_0, \ell'_1, \ell) \in \{0, 1\}$ is 1 exactly when the clockwise rotation of ℓ_0 to ℓ'_1 passes through ℓ [HHK18, Def. 2.5]. Let $z \in H^1(P^*; \mathbb{Z}/4)$ be the class taking the value 1 on a small counterclockwise loop around a corner [HHK18, Def. 3.1].

An immersed circle in P^* is *unobstructed* if every lift of it to the universal cover of P^* is a properly embedded line; by a lemma of Abouzaid [Abo08, Lemma 2.2] this holds if and only if it is homotopically essential and contains no *fishtail*, that is, no subinterval whose endpoints map to the same point and which maps to a nullhomotopic loop [HHK18, Def. 2.1]. A *proper immersion* of an interval is the image of a smooth immersion $I \rightarrow \mathbb{R}^2$ sending the endpoints to lattice points and the interior into $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2$; its slopes at the endpoints are its *limiting slopes* [HHK18, Def. 3.3].

Definition 2.1 ([HHK18, Def. 3.6 and p. 33]). A *restricted immersed circle* is an unobstructed immersed circle $L: S^1 \rightarrow P^*$ with $\mu(L, \ell_1) + z(L) \equiv 0 \pmod{4}$. A *restricted immersed arc* is a proper immersion of an interval whose restriction to the complement of small discs around the

corners contains no fishtail. A *restricted immersed 1-manifold* is a disjoint union of one restricted immersed arc and finitely many restricted immersed circles.

The earring curve L_0 is unobstructed with $\mu(L_0, \ell_1) = 0 = z(L_0)$, so it is a restricted immersed circle [HHK18, pp. 16, 18]. The image of a straight segment joining two lattice points whose interior misses the lattice is a restricted immersed arc [HHK18, p. 18].

Definition 2.2 ([HHK18, Def. 3.11]). A pair (L_0, L_1) of restricted immersed curves is *admissible* if at least one of them is a circle, if L_0 and L_1 intersect transversely, and if, whenever loops α_0 in the domain of L_0 and α_1 in the domain of L_1 have $L_0 \circ \alpha_0$ freely homotopic to $L_1 \circ \alpha_1$ in P^* , both α_0 and α_1 are nullhomotopic.

For an admissible pair, $C(L_0, L_1)$ is the $\mathbb{F}_2$ -vector space on $L_0 \cap L_1$. A *bigon* from p to p' is an orientation-preserving immersion of a disc into P^* with two convex corners, at p and p' , whose boundary, traversed counterclockwise, runs along L_0 from p to p' and then along L_1 from p' back to p ; equivalent bigons are identified [HHK18, Def. 2.10, (5)]. The differential $\partial p = \sum_{p'} \# \mathcal{M}(p, p') p'$ counts bigons from p to p' modulo 2, and by [HHK18, Cor. 3.13] each $\mathcal{M}(p, p')$ contains at most one bigon. A bigon lifts to an immersed disc in $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2$, so its two corners lie on the same component of L_1 ; there is no differential between generators on different components [HHK18, p. 37]. We write $HF(L_0, L_1)$ for the homology.

The complex carries a relative $\mathbb{Z}/4$ -grading $\text{gr}(p, p') = \text{gr}(p) - \text{gr}(p')$, defined for generators on the same component of L_1 [HHK18, Def. 3.7]. The differential lowers it by one, and a bigon from p to p' forces $\text{gr}(p, p') = 1$ [HHK18, Prop. 3.8]. If α_0 is a path in L_0 from p to p' and α_1 a path in L_1 from p' to p , then [HHK18, Lemma 3.10]

$$\text{gr}(p, p') = \mu(\alpha_0, \ell_1) + \mu(\alpha_1, \ell_1) + \tau(T_p L_0, T_p L_1, \ell_1) - \tau(T_{p'} L_0, T_{p'} L_1, \ell_1) + z(\alpha_0 * \alpha_1) \pmod{4}. \quad (2.3)$$

Two consequences are recorded in [HHK18, §§5.1–5.2].

- (1) If L_1 meets the interior of Δ transversely at x , then for small ϵ there are exactly two generators near x : x^+ on the branch of L_0 of slope less than 1 and x^- on the branch of slope greater than 1. They satisfy $\text{gr}(x^+, x^-) = 1$.
- (2) If p_+, p'_+ lie on the branch of slope less than 1 on the front face, α_0 runs along that branch, and L_1 is transverse to Δ at both points, then

$$\text{gr}(p_+, p'_+) = \mu(\alpha_1, \ell_1) + z(\alpha_0 * \alpha_1) \pmod{4}, \quad (2.4)$$

and the same formula holds for two points on the branch of slope greater than 1.

In terms of lifts, $x^\pm$ are the intersections of a lift of L_1 with the lifts $S_\pm^k$ of L_0 near a point of $\mathcal{D}_k$ on the front strip.

Suppose L_1 is a restricted immersed arc with an endpoint at the corner $(0, 0)$ whose limiting slope is bounded away from 1. Then for small ϵ there is a unique intersection point r_+ of L_0 and L_1 converging to that corner as $\epsilon \rightarrow 0$ [HHK18, Lemma 5.1]. When L_1 comes from a knot K as in §2.4, the grading of the component containing the arc is made absolute by

$$\text{gr}(r_+) = \sigma(K) \pmod{4} \quad (2.5)$$

[HHK18, (19)]. Since $\sigma(K)$ is even, (2.5) does not depend on the sign convention for the signature. On circle components Hedden, Herald and Kirk fix the absolute grading by matching one generator with the corresponding generator of Kronheimer and Mrowka's complex [HHK18, §6.1], [KM11b, Prop. 4.4].

Theorem 2.3 ([HHK18, Thm. 5.4]). *Let $L_1: S^1 \rightarrow P^*$ be a restricted immersed circle that misses the left and right edges of the pillowcase, and let $\tilde{L}_1: [0, 2\pi] \rightarrow \mathbb{R}^2$ be a lift of $t \mapsto L_1(e^{it})$.*

Then its vertical degree $d = \frac{1}{2\pi}|\theta(\tilde{L}_1(2\pi)) - \theta(\tilde{L}_1(0))|$ is even, and $HF(L_0, L_1)$ has rank $d/2$ in each of the four gradings.

Remark 2.4. By [HHK18, Thm. 4.1, Cor. 4.6], $HF(L_0, L_1)$ is independent of ϵ and, as a relatively graded group, depends only on the free homotopy classes of the circle components of L_1 and the homotopy class rel endpoints of its arc. The computations in Section 4 are direct and do not use this invariance. Section 5 uses it only through Theorem 2.3, for the circle components: in [HHK18] that theorem is deduced from [HHK18, Cor. 4.6], which follows from [HHK18, Thm. 4.1].

2.4. Tangle decompositions and the conjecture of Hedden, Herald and Kirk. Let $K \subset S^3$ be a knot, decomposed along a Conway sphere as $(S^3, K) = (Y, T) \cup (D, U)$, where (D, U) is the trivial 2-tangle in a ball and the common boundary is identified with $(S^2, \{a, b, c, d\})$ [HHK18, (17)]. Let $R(Y, T)$ be the space of conjugacy classes of traceless $SU(2)$ representations of $\pi_1(Y \setminus T)$, and $R_\pi(Y, T)$ its version perturbed by a holonomy perturbation π supported in the interior of Y . Restriction to the boundary gives $L_1: R_\pi(Y, T) \rightarrow P$, and the earring curve L_0 of §2.2 comes from the trivial side. When L_1 is a restricted immersed 1-manifold and (L_0, L_1) is admissible, Hedden, Herald and Kirk define

$$H^\natural(Y, T, \pi) = HF(L_0, L_1),$$

which is the direct sum of the contributions of the components of $R_\pi(Y, T)$ [HHK18, p. 34, §6.1]. The two abelian traceless representations of $\pi_1(Y \setminus T)$ are the endpoints of the arc component; the one that extends over the knot complement maps to the corner $(0, 0)$, and its perturbation gives the generator r_+ [HHK18, §6.1, §8]. The remaining generators come in pairs $x^\pm$, one pair for each transverse intersection point of L_1 with the interior of Δ . For a torus knot, with the tangle side unperturbed and a small perturbation on the side of the earring, these intersection points correspond to the irreducible traceless representations of the knot group, and there are $1 + |\sigma(K)|$ generators [HHK14, Thm. 11.1], [HHK18, §11.10].

Conjecture 2.5 ([HHK18, Conj. 6.5]). “Given a knot (X, K) in a homology 3-sphere, there exists a 2-tangle decomposition as in Equation (17), such that for suitably small generic perturbations π , $L_1: R_\pi(Y, T) \rightarrow P$ is a restricted immersed 1-manifold and $H^\natural(Y, T, \pi)$ is isomorphic to the reduced instanton homology $I^\natural(X, K)$.”

Their Equation (17) is the decomposition $(X, K) = (Y, T) \cup (D, U)$ along a Conway sphere, which for $X = S^3$ is the decomposition $(S^3, K) = (Y, T) \cup (D, U)$ of this section.

The conjecture asserts the existence of one good decomposition, and the answer does depend on the decomposition: different choices of the gluing data give different curves L_1 [HHK18, p. 55], and with the earring on the other tangle the pillowcase homology of $T(4, 5)$ has the wrong rank [CHKK22, §10.3]. Hedden, Herald and Kirk verify the conjecture for all two-bridge knots [HHK18, §7]. For the torus knots $T(3, n)$ with $n \geq 4$ prime to 3, suitable decompositions from [HHK18, §§10–11] satisfy it, with the $\mathbb{Z}/4$ -gradings [Wue26c]. Section 4 obtains the decompositions used for $P(-2, 3, q)$ from these by regluing.

2.5. The knots. We use the convention that positive knots have negative signature, as do [DS24a, DS24b, Wue26c] and the tables of [HHK18]; for instance $\sigma(T(3, 5)) = -8$ [HHK18, p. 60]. Braid generators are written σ_1, σ_2 and are positive crossings, so that the closure of $(\sigma_1 \sigma_2)^n$ is the positive torus knot $T(3, n)$.

For odd $q \geq 1$, $K_q = P(-2, 3, q)$ is the pretzel knot with three twist regions of $-2, 3$ and q half-twists, with the chirality for which its first three members are the positive torus knots $P(-2, 3, 1) = T(2, 5)$, $P(-2, 3, 3) = T(3, 4)$ and $P(-2, 3, 5) = T(3, 5)$; for $q \geq 7$ it is hyperbolic

(Proposition 4.5). Section 4 decomposes it by regluing Hedden, Herald and Kirk's decomposition of $T(3, 5)$ (Theorem 4.4).

For $n \geq 7$ with $n \equiv 1, 2 \pmod{6}$ and $m \geq 1$, the twisted torus knot $T(3, n; 2, m)$ is the closure of the three-strand braid

$$(\sigma_1 \sigma_2)^n \sigma_1^{2m},$$

that is, $T(3, n)$ with m positive full twists added on two adjacent strands. It is a positive braid closure. Section 5 treats these knots with decompositions obtained from those of $T(3, n)$ in the same way.

3. THE INSTANTON SIDE

Throughout this section $K_q = P(-2, 3, q)$ with $q \geq 3$ odd. For a knot K let $\ell(K) = \sum_i |a_i(\Delta_K)|$ be the coefficient norm of its Alexander polynomial. Kronheimer and Mrowka's integral spectral sequence from reduced Khovanov homology, their rational identification of $I^\natural$ with KHI , and the Alexander-graded Euler characteristic of KHI give

$$\ell(K) \leq \dim_{\mathbb{Q}} I^\natural(K; \mathbb{Q}) \leq \dim_{\mathbb{Q}} \widetilde{Kh}(K; \mathbb{Q}) \quad (3.1)$$

[KM11b, Props. 1.2 and 1.4], [KM10, Thm. 1.1 and Cor. 1.2]. We prove Theorem 1.1 by showing that both bounds equal $q + 2$ and that the spectral sequence collapses over $\mathbb{Z}$. We then record the signature of K_q and the $\mathbb{Z}/4$ -graded dimensions of $I^\natural(K_q)$ that Section 4 compares with the pillowcase.

3.1. The Alexander polynomial. For $m \geq 1$ write $[m]_x = 1 + x + \cdots + x^{m-1}$.

Lemma 3.1. *For every odd $q \geq 3$ put*

$$Q_q(x) = (x - 2)[2]_x[3]_x[q]_x + [2]_x[3]_x + [2]_x[q]_x + [3]_x[q]_x.$$

Then

$$\Delta_{K_q}(t) \doteq Q_q(-t) = 1 - t + \sum_{j=3}^q (-1)^{j+1} t^j - t^{q+2} + t^{q+3}, \quad (3.2)$$

where $\doteq$ denotes equality up to a unit $\pm t^k$. In particular $\ell(K_q) = q + 2$ and $\det K_q = |\Delta_{K_q}(-1)| = |q - 6|$.

Proof. Hironaka proves that the Alexander polynomial of the $(2, 3, q, -1)$ -pretzel knot is $Q_q(-t)$, with Q_q the first displayed expression [Hir01, Thm. 1.2]; the erratum [Hir02] corrects the statement for pretzel links with more than three positive entries and leaves the three-entry case used here unchanged. The $(2, 3, q, -1)$ -pretzel knot is K_q , because the rational tangles of fractions $1/2$ and -1 sum to the rational tangle of fraction $-1/2$; Hironaka records this identification for $q = 7$ [Hir01, p. 442]. Expanding the geometric sums gives $Q_q(x) = 1 + x - \sum_{j=3}^q x^j + x^{q+2} + x^{q+3}$, and substituting $x = -t$ gives the right-hand side of (3.2), since q is odd. It has four coefficients of absolute value 1 outside the sum and $q - 2$ inside, so $\ell(K_q) = q + 2$. At $t = -1$ the terms outside the sum contribute $2 + 2$ and each of the $q - 2$ terms inside contributes -1 , so $\Delta_{K_q}(-1) = 6 - q$. $\square$

Remark 3.2. At $q = 7$, $Q_7(x) = 1 + x - x^3 - x^4 - x^5 - x^6 - x^7 + x^9 + x^{10}$ is Lehmer's polynomial. The substitution $t \mapsto -t$ in (3.2) is part of Hironaka's formula, not a normalization ambiguity of the Alexander polynomial. The same polynomial is obtained from the reduced Burau representation of the braid $(\sigma_1 \sigma_2)^5 \sigma_1^{q-5}$, whose closure is K_q for $q \geq 5$ (Section 4).

The following facts are used in Section 4 to show that K_q is hyperbolic and not two-bridge.

Lemma 3.3. *For every odd $q \geq 3$ the knot K_q is neither quasi-alternating nor alternating, and hence it is not two-bridge. For $q \geq 7$ it is not a torus knot.*

Proof. For a quasi-alternating knot K the rank of $I^\natural(K)$ equals $|\det K|$ [KM11b, Cor. 1.6]. By (3.1) and Lemma 3.1 the rank of $I^\natural(K_q)$ is at least $q + 2$, whereas $|\det K_q| = |q - 6| < q + 2$. For an alternating knot the coefficients of the Alexander polynomial alternate in sign [Cro59, Mur58], so that $|\Delta_K(-1)| = \ell(K)$; for K_q the two sides are $|q - 6|$ and $q + 2$. Two-bridge knots are alternating.

For the last assertion, recall that for coprime $1 < a < b$ the Alexander polynomial $\Delta_{T(a,b)} = (1-t)(1-t^{ab})/((1-t^a)(1-t^b))$ begins $1-t+t^a-\dots$ and has degree $(a-1)(b-1)$; it is unchanged under mirroring. In (3.2) the coefficient of t^2 is 0 and that of t^3 is 1, so a torus knot with the same Alexander polynomial would have $a = 3$, and comparing degrees gives $2(b-1) = q + 3$, so $b = (q+5)/2 \geq 6$. For $b \geq 6$ the coefficient of t^5 in $\Delta_{T(3,b)} = (1-t)(1+t^3+t^6+\dots)(1+t^b+\dots)(1-t^{3b})$ is 0, while the coefficient of t^5 in (3.2) is 1. $\square$

3.2. Reduced Khovanov homology and the integral spectral sequence. Manion computes the reduced Khovanov homology over $\mathbb{Z}$ of the pretzel knots $P(-p, r, s)$ with $p, r, s > 0$ and $2 \leq p \leq r, s$ that are not quasi-alternating, and proves that it is free [Man18, Thm. 1.1 and Thm. 3.3]. For p even it is $\mathbb{Z}^{p^2} \oplus \mathbb{Z}^{(r-p)(s-p)}$, supported in two adjacent δ -gradings. By Lemma 3.3, K_q satisfies these hypotheses for every odd $q \geq 3$, and substituting $(p, r, s) = (2, 3, q)$ gives

$$\widetilde{Kh}(K_q; \mathbb{Z}) \text{ is free of rank } 4 + (3-2)(q-2) = q+2. \quad (3.3)$$

The reduced complex of the mirror is the integral dual complex, so by the universal coefficient theorem the reduced Khovanov homology of the mirror is also free of rank $q + 2$.

Proof of Theorem 1.1. Kronheimer and Mrowka construct a spectral sequence of abelian groups whose E_2 page is the reduced Khovanov homology of the mirror and whose abutment is $I^\natural(K_q; \mathbb{Z})$ [KM11b, Prop. 1.2 and Thm. 8.12]; the underlying filtration, indexed by the vertices of a cube of resolutions, has finite length. After tensoring with $\mathbb{Q}$, (3.3) gives $\dim_{\mathbb{Q}} I^\natural(K_q; \mathbb{Q}) \leq q + 2$. Kronheimer and Mrowka also identify $I^\natural(K_q; \mathbb{Q})$ with $KHI(K_q; \mathbb{Q})$ [KM11b, Prop. 1.4]; the Alexander-graded Euler characteristic of KHI is the Alexander polynomial up to sign, so the dimension of KHI is at least $\ell(K_q)$ [KM10, Thm. 1.1 and Cor. 1.2]. The identification of [KM11b, Prop. 1.4] is stated over $\mathbb{Q}$; the statements of [KM10] are made over $\mathbb{C}$ and pass to $\mathbb{Q}$ by flat base change. By Lemma 3.1, $\dim_{\mathbb{Q}} I^\natural(K_q; \mathbb{Q}) \geq \ell(K_q) = q + 2$, so the two rational bounds agree. Total dimension cannot increase from one page to the next in a spectral sequence of finite-dimensional vector spaces, so every differential vanishes after tensoring with $\mathbb{Q}$.

The E_2 page of the integral spectral sequence is free of rank $q + 2$. Inductively each page is free, and a homomorphism between free abelian groups whose rationalization vanishes is zero; hence every integral differential vanishes. The associated graded group of $I^\natural(K_q; \mathbb{Z})$ is therefore free of rank $q + 2$, and its finite filtration splits because extensions of free abelian groups split. This proves $I^\natural(K_q; \mathbb{Z}) \cong \mathbb{Z}^{q+2}$, and the assertions over $\mathbb{Q}$ and $\mathbb{F}_2$ follow by base change and the universal coefficient theorem. $\square$

Remark 3.4. The argument with the two-sided rank bound (3.1) over $\mathbb{Q}$ is Lobb and Zentner's proof that the Kronheimer–Mrowka spectral sequence of K_q is trivial [LZ20, Prop. 4.7]; their $I^\natural$ is Kronheimer and Mrowka's $I^\natural$ of the mirror, and the rank is unchanged by mirroring. Smith uses the same argument for $q = 5$ [Smi24, proof of Thm. 5.9]. The rational rank also follows from the work of Li and Ye: K_q is an instanton L-space knot [LY21, Prop. 7.1(2)], so each Alexander-graded summand of $KHI(K_q)$ has dimension at most one [LY21, Thm. 1.9], and the dimension of KHI is then the number of nonzero coefficients of the Alexander polynomial [KM10, Thm. 1.1]. The freeness in Theorem 1.1 additionally uses Manion's integral calculation. For $q \geq 7$ it also follows from the work of Daemi and Scaduto (Proposition 3.6 below). For the torus knots $K_3 = T(3, 4)$ and $K_5 = T(3, 5)$ it can be read off from the integral S-complexes that

Daemi and Scaduto write down [DS24a, §§9.3–9.4]: there the differential has rank one, because the rank of $I^\natural$ is known, and one of its components counts a single instanton, so the homology is torsion free. Finally, $\det K_q = |q - 6|$ never vanishes and equals 1 exactly for $q = 5, 7$, the two members whose double branched cover is an integral homology sphere.

3.3. The signature. We use the convention that positive knots have negative signature, so that $\sigma(T(3, 4)) = -6$ and $\sigma(T(3, 5)) = -8$; this is the convention of [HHK18] and [Wue26c].

Proposition 3.5. *For odd $q \geq 1$,*

$$\sigma(K_q) = \begin{cases} -(q+3), & q \in \{1, 3, 5\}, \\ -(q+1), & q \geq 7. \end{cases}$$

Proof. This is [DS24b, Lemma 8.29], stated there in the same convention; Daemi and Scaduto argue by induction from $K_1 = T(2, 5)$, using the congruence $|\sigma(K)| + 1 \equiv \det K \pmod{4}$ and the change of the signature under a crossing change between links of nonzero determinant. An independent check is the formula of Gordon and Litherland [GoLi78, Thm. 6], applied to the checkerboard surface of the standard diagram: two discs joined by three twisted bands with -2 , 3 and q half-twists. With respect to the two white regions between the bands, its Goeritz form is $\eta \begin{pmatrix} 1 & -3 \\ -3 & q+3 \end{pmatrix}$ for a sign η fixed by the choice of coloring, with determinant $q - 6$. The strands of K_q run antiparallel through the band with -2 half-twists and parallel through the other two, so the Gordon–Litherland correction term is $\mu = \eta(q+3)$, one contribution from each of the $3+q$ crossings of the two odd bands. The form has signature 2η for $q \geq 7$ and 0 for $q \leq 5$, so $\sigma(K_q) = -\eta(q+1)$ for $q \geq 7$ and $-\eta(q+3)$ for $q \leq 5$. The value $\sigma(K_5) = \sigma(T(3, 5)) = -8$ fixes $\eta = 1$. $\square$

3.4. The $\mathbb{Z}/4$ -graded instanton homology. Daemi and Scaduto associate to a knot K an S-complex

$$\tilde{C}(K) = C(K) \oplus C(K)[-1] \oplus \mathbb{Z}_{(0)}, \quad C[i]_j = C_{i+j}, \quad (3.4)$$

a decomposition of $\mathbb{Z}/4$ -graded modules in which $C(K)$ is generated by the irreducible critical points of the singular Chern–Simons functional and the summand $\mathbb{Z}_{(0)}$, in degree 0, by the reducible one. The differential has the shape

$$\tilde{d} = \begin{pmatrix} d & 0 & 0 \\ v & -d & \delta_2 \\ \delta_1 & 0 & 0 \end{pmatrix},$$

where δ_1 and δ_2 count instantons to and from the reducible [DS24b, Def. 2.1 and (1.9)], [DS24a, Def. 3.21]. The S-complex structure is the map $\chi: \tilde{C}(K) \rightarrow \tilde{C}(K)$ of degree $+1$ given in the decomposition (3.4) by the matrix $\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, which maps $C(K)$ identically onto $C(K)[-1]$; it satisfies $\chi^2 = 0$ and $\chi\tilde{d} + \tilde{d}\chi = 0$ [DS24a, §4.1]. The homology of $(\tilde{C}(K), \tilde{d})$ is $I^\natural(K)$: there is a chain homotopy equivalence between Kronheimer and Mrowka’s complex and $\tilde{C}(K)$, homogeneous for the $\mathbb{Z}/4$ -gradings, and for knots in S^3 the resulting isomorphism on homology has degree $\sigma(K)$ modulo 4 [DS24a, Thm. 8.9]. We write $I(K)$ for the homology of $(C(K), d)$, the irreducible instanton homology.

Proposition 3.6. *Let $q \geq 7$ be odd and put $N_1 = \lceil \frac{q+1}{4} \rceil$ and $N_3 = \lfloor \frac{q+1}{4} \rfloor$. Then $I^\natural(K_q; \mathbb{Z})$ is free of rank $q+2$, and in Kronheimer and Mrowka’s absolute grading its graded ranks are*

$$(1 + N_3, N_1, N_1, N_3) \quad \text{in degrees } \text{gr} - \sigma(K_q) = 0, 1, 2, 3. \quad (3.5)$$

The same holds with coefficients in $\mathbb{Q}$ and in $\mathbb{F}_2$.

Proof. For $q \geq 7$, Daemi and Scaduto prove $I(K_q) \cong \mathbb{Z}_{(1)}^{N_1} \oplus \mathbb{Z}_{(3)}^{N_3}$, with the subscripts denoting $\mathbb{Z}/4$ degrees in the grading of (3.4) [DS24b, Prop. 8.30]; the degrees come from [DS24b, Prop. 8.28] and the freeness from [DS24b, Rem. 8.27]. The filtration $C[-1] \subset C[-1] \oplus \mathbb{Z}_{(0)} \subset \tilde{C}$ induces a spectral sequence

$$I(K_q) \oplus I(K_q)[-1] \oplus \mathbb{Z}_{(0)} \implies I^\natural(K_q; \mathbb{Z}) \quad (3.6)$$

[DS24b, (8.5)]. Daemi and Scaduto state the homology groups of their §8 as $\mathbb{Z}/2$ -graded, but the spectral sequence respects the $\mathbb{Z}/4$ -grading: by the shape of $\tilde{d}$, the summands $C[-1]$ and $C[-1] \oplus \mathbb{Z}_{(0)}$ are subcomplexes, and (3.4) is a decomposition of $\mathbb{Z}/4$ -graded modules, so (3.6) is a spectral sequence of $\mathbb{Z}/4$ -graded groups. Since a class of degree j in I sits in degree $j+1$ in $I[-1]$, the initial page has ranks $1 + N_3$, N_1 , N_1 and N_3 in degrees 0, 1, 2 and 3, for a total of $1 + 2(N_1 + N_3) = q + 2$.

By Theorem 1.1, the rank of the abutment is also $q + 2$. Hence every differential vanishes after tensoring with $\mathbb{Q}$, and, as in the proof of Theorem 1.1, every integral differential vanishes and the filtration of $I^\natural(K_q; \mathbb{Z})$ splits. So $I^\natural(K_q; \mathbb{Z})$ is free with the graded ranks of the initial page in the grading of (3.4). By [DS24a, Thm. 8.9], Kronheimer and Mrowka's absolute grading is this grading shifted by $\sigma(K_q)$; since $\sigma(K_q)$ is even, the sign of the shift does not matter modulo 4. The assertions over $\mathbb{Q}$ and $\mathbb{F}_2$ follow by base change. $\square$

For $q = 7, 9, 11$ and 13 the vector (3.5) is $(3, 2, 2, 2)$, $(3, 3, 3, 2)$, $(4, 3, 3, 3)$ and $(4, 4, 4, 3)$. For $q = 3, 5$ the same spectral sequence has initial rank $q + 4$ and a differential of rank one. The graded ranks of $I^\natural$ of the torus knots $K_3 = T(3, 4)$ and $K_5 = T(3, 5)$ are $(1, 1, 2, 1)$ and $(2, 1, 2, 2)$ in degrees $\text{gr} - \sigma = 0, 1, 2, 3$ [Wue26c, (2.5) and Table 2], and these are the vectors with which the pillowcase homology of the two torus knots is compared in [Wue26c]. They are instances of the graded formula of [Wue26b], which determines $I^\natural(K; \mathbb{Q})$ for every torus knot K , and the S-complexes of the two knots are described in [Wue26a].

4. PILLOWCASE HOMOLOGY OF THE PRETZEL KNOTS

In this section we prove Theorem 1.2. The decomposition is the one used by Hedden, Herald and Kirk for $T(3, 5)$ [HHK18, §10], reglued along the Conway sphere by $m = (q - 5)/2$ Dehn twists. The proof has four steps.

- (1) The reglued knot is $K_q = P(-2, 3, q)$ (§4.1). The argument uses only the fundamental group and the signature and, apart from the base case (Remark 4.3), no planar diagram of the decomposition.
- (2) On the pillowcase the regluing replaces L_1 by its image under a linear shear (§4.2).
- (3) The unperturbed variety of the $T(3, 5)$ tangle maps onto two straight polygons (§4.3). This rests on a word in $\pi_1(Y \setminus T)$ that is trivial on the whole central circle of the variety.
- (4) The perturbed curves are those of [Wue26c], sheared (§4.4). From them we compute the generators, the gradings and the bigons by hand (§§4.5–4.7), for both ways in which the perturbation can resolve the double points.

No invariance theorem of [HHK18] is used. As in [Wue26c], results of [HHK18] are cited by their numbering in arXiv:1501.00028v1.

Throughout, $\varepsilon = (\varepsilon_A, \varepsilon_B)$ denotes the parameters of the holonomy perturbation on the tangle side ((P1) in §4.3), and $\epsilon > 0$ the earring parameter of §2.2. As in [Wue26c, §6], ε is fixed first and ϵ is then taken sufficiently small, and all statements about intersection points with L_0 hold for every sufficiently small ϵ .

4.1. The decomposition.

The tangle. Let (Y, T) be the tangle of [HHK18, §10] for $T(3, 5)$, with the integers $(r, s) = (2, -1)$ satisfying $3r + 5s = 1$; this is the tangle of [Wue26c, §3] for $n = 5$. The group $\pi_1(Y \setminus T)$ is free on two elements A, B (the elements A_1, B_1 of [HHK18, §10], called A, B in [HHK14, §11]). At $\varepsilon = 0$ the meridians of the four boundary punctures are

$$a = A^2 B^3, \quad b = B^{-2} A^{-1}, \quad c = B^{-2} a B^2, \quad (4.1)$$

by [HHK14, (42)], where Hedden, Herald and Kirk's torus-knot parameters are 3 and 5 and $(r, s) = (2, -1)$, so that the four exponents in their formula are $s + 3 = 2$, $5 - r = 3$, $r = 2$ and $s = -1$ (see [Wue26c, (3.2)]), and the fourth meridian d satisfies $ba = cd$ [HHK14, §3.3]. More generally, for $n = 3\nu + 2$ with $\nu \geq 1$, the tangle of [HHK18, §10] for $T(3, n)$ with $(r, s) = (\nu + 1, -1)$ has

$$a = A^2 B^{2\nu+1}, \quad b = B^{-(\nu+1)} A^{-1}, \quad c = B^{-(\nu+1)} a B^{\nu+1}. \quad (4.2)$$

The trivial tangle and the twist. Let (D, U) be the trivial tangle of [HHK18, (17)]: D is the unit ball, and U consists of the two vertical arcs $\{(\pm 1/\sqrt{2}, 0, t) : |t| \leq 1/\sqrt{2}\}$, one joining the punctures a, d and the other joining b, c . The group $\pi_1(D \setminus U)$ is free, and its boundary meridians satisfy $a = d$ and $b = c$ [HHK14, Prop. 6.1], [FKP17, p. 4]. Let $e \subset \partial D$ be the equator $\{z = 0\}$; it separates $\{a, b\}$ from $\{c, d\}$ and bounds the horizontal disc $D_e = D \cap \{z = 0\}$, which meets each arc of U once. Let t_e be the Dehn twist along e that acts on $\pi_1(S^2 \setminus \{a, b, c, d\}) = \langle a, b, c, d \mid ba = cd \rangle$, with base point on the side of c, d , by

$$t_e(a) = (ba) a (ba)^{-1}, \quad t_e(b) = (ba) b (ba)^{-1}, \quad t_e(c) = c, \quad t_e(d) = d. \quad (4.3)$$

It fixes $ba = cd$. Hedden, Herald and Kirk use the half twist g in the hemisphere containing c and d that exchanges c and d , and note that g^2 is the Dehn twist about e [HHK18, proof of Thm. 9.2, p. 49]. With base point near a , g_* fixes a, b and sends $c \mapsto d$, $d \mapsto d^{-1}cd$. So g_*^2 conjugates c and d by $(cd)^{-1} = (ba)^{-1}$ and fixes a, b , and composing $(g_*^2)^{-1}$ with conjugation by $(ba)^{-1}$ gives t_e^{-1} . Thus t_e and g^2 are the same mapping class.

Recovering a knot from (Y, T) requires a choice of identification of ∂D with ∂Y [HHK18, after (17)]. For $h \in \mathbb{Z}$ let

$$K^{(h)} = (Y, T) \cup_{t_e^h} (D, U)$$

be the knot obtained by precomposing the identification of [HHK18, (17)], on the side of Y , with t_e^h : a boundary loop x of D is identified with the loop $t_e^h(x)$ of Y .

Lemma 4.1. *For every $h \in \mathbb{Z}$, $K^{(h)}$ is a knot in S^3 and*

$$\pi_1(S^3 \setminus K^{(h)}) \cong \langle A, B \mid t_e^h(b) = c \rangle, \quad t_e^h(b) = (ba)^h b (ba)^{-h},$$

with a, b, c the words of the tangle. The knots $K^{(h)}$ and $K^{(h-1)}$ differ by one full twist of the two arcs of U about each other along D_e ; in particular they are related by one crossing change.

Proof. Gluing two balls along their boundary gives S^3 , and t_e fixes each puncture, so the strands are connected as in $K^{(0)}$, which is a knot. By van Kampen's theorem, $\pi_1(S^3 \setminus K^{(h)})$ is the amalgam of $\pi_1(Y \setminus T) = F(A, B)$ and $\pi_1(D \setminus U)$, free on $x = a = d$ and $y = b = c$, over the four-punctured sphere. Eliminating x and y leaves the relations $t_e^h(b) = t_e^h(c) = c$ and $t_e^h(a) = t_e^h(d) = d$. The second follows from the first, because t_e^h fixes $ba = cd$: $t_e^h(a) = t_e^h(b)^{-1}ba = c^{-1}ba = d$. Regluing by t_e^{-1} amounts to replacing U by its image under a twist map supported in a collar of ∂D , which is U with one full twist of its two arcs along D_e . Changing one of the two crossings of a full twist of two strands undoes it. $\square$

At $h = 0$ the relation $b = c$ reads $B^{-2}A^{-1} = B^{-2}A^2B^5$, that is $A^3B^5 = 1$, the group of $T(3, 5)$, as it must be.

Clay and Watson define the twisted torus knots $T_{3,n}^m$ by adding m positive full twists on two of the three strands of $T(3, n)$ [CW13, §3.2]; $T_{3,n}^m$ is the closure of the braid $(\sigma_1\sigma_2)^n\sigma_1^{2m}$ (§2.5). For $n = 3\nu + 2$ they give the presentation [CW13, Prop. 22]

$$\pi_1(S^3 \setminus T_{3,n}^m) \cong \langle \alpha, \beta \mid \alpha^2(\beta^{-\nu}\alpha)^m\alpha = \beta^{2\nu+1}(\beta^{-\nu}\alpha)^m\beta^{\nu+1} \rangle, \quad m \geq 0. \quad (4.4)$$

Lemma 4.2. *Let $n = 3\nu + 2$ with $\nu \geq 1$, let (Y, T) be the tangle with the words (4.2), and put $C = AB^\nu$. For every $m \geq 0$ the relation $t_e^{-m}(b) = c$ is equivalent to the vanishing of the cyclic word*

$$R_{n,m} = B^{\nu+1}A C^m A^2 B^{2\nu+1} C^{-m},$$

and under $\alpha = A$, $\beta = B^{-1}$ the relator of (4.4) is a cyclic permutation of $R_{n,m}$. In particular $\pi_1(S^3 \setminus K^{(-m)}) \cong \pi_1(S^3 \setminus T_{3,n}^m)$.

Proof. Since $ba = B^{-(\nu+1)}AB^{2\nu+1} = B^{-(\nu+1)}CB^{\nu+1}$, we have $(ba)^j = B^{-(\nu+1)}C^jB^{\nu+1}$. Hence $t_e^{-m}(b) = B^{-(\nu+1)}C^{-m}A^{-1}B^{-(\nu+1)}C^mB^{\nu+1}$, and $t_e^{-m}(b) = c$ becomes $C^{-m}A^{-1}B^{-(\nu+1)}C^m = A^2B^{2\nu+1}$. The inverse of the corresponding relator is $A^2B^{2\nu+1}C^{-m}B^{\nu+1}AC^m$, a cyclic permutation of $R_{n,m}$. On the other side, substitute $\alpha = A$ and $\beta = B^{-1}$ in the relator of (4.4):

$$\alpha^2(\beta^{-\nu}\alpha)^m\alpha\beta^{-(\nu+1)}(\beta^{-\nu}\alpha)^{-m}\beta^{-(2\nu+1)} \mapsto A^2(B^\nu A)^m A B^{\nu+1}(B^\nu A)^{-m}B^{2\nu+1}.$$

Using $A(B^\nu A)^m = C^m A$ and $(B^\nu A)^{-m} = B^\nu C^{-m} B^{-\nu}$, this is $AC^m A^2 B^{2\nu+1} C^{-m} B^{\nu+1}$, again a cyclic permutation of $R_{n,m}$. $\square$

The group of a knot does not distinguish it from its mirror image, so the chirality of the base case enters as a premise.

Remark 4.3 (Base case). The decomposition [HHK18, (17)] gives the positive torus knot $T(3, 5)$, of signature -8 . Hedden, Herald and Kirk call $K^{(0)}$ the $(3, 5)$ torus knot without fixing its handedness [HHK18, §10], and their comparison in [HHK18, §11] uses σ only modulo 4, so it does not decide the question; the knot group cannot detect it either. We verify it by a computer-assisted argument. From a planar diagram of [HHK18, Fig. 18], SnapPy [CDGW], with verified computation of canonical cell decompositions by interval arithmetic [HIK+16, DHL15], shows that the complement of the link $K^{(0)} \cup e$ is isometric to the complement of the closure of the positive braid $(\sigma_1\sigma_2)^5$ together with an unknotted circle encircling two of its strands, and that every such isometry preserves the orientation and the meridians (Appendix A). Such an isometry extends to an orientation-preserving homeomorphism of S^3 carrying $K^{(0)}$ to the closure of $(\sigma_1\sigma_2)^5$, the positive torus knot. The identification of the knots below and the graded comparison in Theorem 1.2 rest on this.

Theorem 4.4. *For every $m \geq 1$, the knot $K^{(-m)}$ is the pretzel knot $K_{5+2m} = P(-2, 3, 5 + 2m)$, the closure of the positive braid $(\sigma_1\sigma_2)^5\sigma_1^{2m}$, and $\sigma(K^{(-m)}) = -(2m + 6)$.*

Proof. By Lemma 4.2 with $\nu = 1$, the groups of $K^{(-m)}$ and of $T_{3,5}^m$ are isomorphic. Clay and Watson state that $T_{3,5}^m$ is $P(-2, 3, 5 + 2m)$ [CW13, §3.2]; for $m = 1$ this is also stated in [DS24b, p. 164]. We use this only up to mirror image. In particular both knots are nontrivial, since the Alexander polynomial of their common group is that of K_{5+2m} , which is not constant (Lemma 3.1). Both groups are generated by two elements, so both knots are prime by Norwood's theorem [Nor82]. Prime knots with isomorphic groups are equivalent, by a homeomorphism of S^3 that may reverse orientation [GoLu89, Cor. 2.1]. Hence $K^{(-m)}$ is K_{5+2m} or its mirror image.

To fix the chirality, recall $\sigma(K_q) = -(q + 1)$ for $q \geq 7$ and $\sigma(K_5) = \sigma(T(3, 5)) = -8$ (Proposition 3.5); the mirror image has the opposite signature. By Lemma 4.1, $K^{(-m-1)}$ and $K^{(-m)}$ are related by one crossing change, so their signatures differ by at most 2 [Mur65, proof

of Thm. 10.1, (10.2)]. Suppose $\sigma(K^{(-m)}) = -s_m$ with $s_0 = 8$ and $s_m = 2m + 6$ for $m \geq 1$; for $m = 0$ this is Remark 4.3. The two candidates for $K^{(-m-1)}$ have signature $\pm(2m + 8)$, and only $-(2m + 8)$ lies within 2 of $-s_m$. Hence $K^{(-m-1)} = K_{2m+7}$, and induction on m shows $K^{(-m)} = K_{5+2m}$ for every $m \geq 1$.

It remains to see that K_{5+2m} , rather than its mirror image, is the closure of $(\sigma_1\sigma_2)^5\sigma_1^{2m}$. By Clay and Watson, used up to mirror image, this closure is K_{5+2m} or its mirror image. Changing one crossing of σ_1^{2m} from positive to negative gives $(\sigma_1\sigma_2)^5\sigma_1^{2m-2}$, and for knots $K_{\pm}$ related by such a change $\sigma(K_+) - \sigma(K_-) \in \{0, -2\}$ in our convention (see the proof of Proposition 5.5). By induction from $\sigma(T(3, 5)) = -8$, the closure has negative signature. Since $\sigma(K_{5+2m}) = -(2m + 6) < 0$ and its mirror image has positive signature, the closure is K_{5+2m} . $\square$

The same argument applies to every $n \equiv 2 \pmod{3}$ once the signature of $T(3, n; 2, m)$ is known; this is used in Section 5.

Proposition 4.5. *For odd $q \geq 7$ the knot K_q is hyperbolic and is not a two-bridge knot.*

Proof. By Lemma 3.3, K_q is neither two-bridge nor a torus knot. It is the Montesinos knot with rational tangles $-1/2, 1/3, 1/q$, all of denominator at least 2. By Oertel's theorem [Oer84, Cor. 5], the complement of such a Montesinos link carries a complete hyperbolic structure unless the link is a torus link or one of the links $K(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2})$, $K(\frac{2}{3}, -\frac{1}{3}, -\frac{1}{3})$, $K(\frac{1}{2}, -\frac{1}{4}, -\frac{1}{4})$, $K(\frac{1}{2}, -\frac{1}{3}, -\frac{1}{6})$ or a mirror image of one of these. For each of the four listed links the sum of the fractions is 0. The determinant of a Montesinos link with fractions β_i/α_i is $|\alpha_1 \cdots \alpha_r \sum_i \beta_i/\alpha_i|$, the order of the first homology of its double branched cover (infinite, and the determinant 0, when the sum vanishes), whereas the determinant of a knot is odd; so none of them, and none of their mirror images, is a knot. Conventions for the sign of a rational tangle differ between authors, which may replace K_q by its mirror image; the list of exceptions is closed under mirror image. Hence K_q is hyperbolic. $\square$

Oertel's corollary does not list the torus links, which is why Lemma 3.3 is needed. Explicitly, apart from two-bridge knots the only non-hyperbolic Montesinos knots are $P(-2, 3, 3) = T(3, 4)$ and $P(-2, 3, 5) = T(3, 5)$ and their mirror images [IJ09, Rem. 1], [Mei14, p. 440]. Futer and Guéritaud give a published proof of the classification of non-hyperbolic arborescent links, which they attribute to Bonahon and Siebenmann [FG09, Thm. 1.5].

4.2. The twist acts by a shear. For $h \in \mathbb{Z}$ let $S_h: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear map $S_h(\gamma, \theta) = (\gamma, \theta - 2h\gamma)$.

Lemma 4.6. *Let (Y, T) be a 2-tangle as in [HHK18, (17)], with a holonomy perturbation supported in the interior of Y , and let $h \in \mathbb{Z}$. For the decomposition $(Y, T) \cup_{t_h} (D, U)$:*

- (a) $R_{\pi}(Y, T)$ is unchanged, its restriction map is $S_h \circ L_1$, where L_1 is the restriction map for the identification of [HHK18, (17)], and L_0 is unchanged;
- (b) S_h descends to a homeomorphism of P fixing each corner and to an orientation-preserving diffeomorphism of P^* ; it preserves γ , hence the open strip $\{0 < \gamma < \pi\}$, the two edges, and the vertical degree of closed curves that miss the edges;
- (c) S_h maps proper immersions of intervals to proper immersions, adding $-2h$ to both limiting slopes, and preserves embeddedness of lifts and the absence of fishtails;
- (d) for closed curves S_h preserves z and $\mu(\cdot, \ell_1)$; hence it maps restricted immersed 1-manifolds to restricted immersed 1-manifolds;
- (e) the corner images of the two abelian representations are unchanged;
- (f) for every immersed path α , $\mu(S_h \circ \alpha, \ell_1) = \mu(\alpha, S_h^{-1}\ell_1)$, where $S_h^{-1}\ell_1$ is the constant line field spanned by $(1, 1 + 2h)$.

Proof. (a) The perturbation is supported in the interior of Y , so $R_\pi(Y, T)$ does not change. Let a representation restrict to $a \mapsto \mathbf{i}$, $b \mapsto e^{\gamma \mathbf{k}} \mathbf{i}$, $c \mapsto e^{\theta \mathbf{k}} \mathbf{i}$ on ∂Y . Then $\rho(ba) = -e^{\gamma \mathbf{k}}$, so by (4.3), $\rho(t_e^h a) = e^{2h\gamma \mathbf{k}} \mathbf{i}$, $\rho(t_e^h b) = e^{(2h+1)\gamma \mathbf{k}} \mathbf{i}$, and c is fixed. Conjugating by $e^{-h\gamma \mathbf{k}}$ gives $(\mathbf{i}, e^{\gamma \mathbf{k}} \mathbf{i}, e^{(\theta-2h\gamma) \mathbf{k}} \mathbf{i})$, the point $S_h(\gamma, \theta)$, with $\gamma \in [0, \pi]$ unchanged. The trivial tangle keeps the identification of [HHK18, (17)], so L_0 is unchanged.

(b) S_h is linear of determinant 1 and commutes with -1 ; it sends $(2\pi, 0)$ to $(2\pi, -4\pi h)$ and fixes $(0, 2\pi)$, so it normalizes G . It maps $(\pi a, \pi b)$ to $(\pi a, \pi(b - 2ha)) \equiv (\pi a, \pi b)$ modulo $2\pi\mathbb{Z}^2$.

(c) Embeddedness of lifts and the absence of fishtails are invariant under the homeomorphism S_h of P^* .

(d) S_h fixes the corners and preserves orientation, so it preserves z . For a closed curve, $\mu(\cdot, \ell)$ is the total rotation of the tangent line divided by π , for every constant line field ℓ , because the deck transformations have derivative $\pm I$. Along the path $s \mapsto S_{sh}$, $0 \leq s \leq 1$, this total rotation is a continuous integer, hence constant. The rest follows from (b), (c) and Definition 2.1.

(e) This follows from (b).

(f) The tangent line of $S_h \circ \alpha$ equals ℓ_1 exactly when the tangent line of α equals $S_h^{-1} \ell_1$. Since S_h preserves orientation, the senses of passage agree, and the endpoint conventions correspond. $\square$

The computation in (a) is that of [HHK18, (28)]: there the half twist g acts on the pillowcase by $(\gamma, \theta) \mapsto (\gamma, \theta - \gamma)$, and [HHK18, (29)–(30)] obtain the action of $\mathrm{PSL}_2(\mathbb{Z})$ from g and a 3-cycle. Since $t_e = g^2$, t_e^h acts by S_h . Our additions are the observation that regluing by t_e^h is a choice of identification permitted by [HHK18, (17)], and properties (b)–(f). The action of the mapping class group on the pillowcase is also [CHK22, Thm. B], and its counterpart for immersed curves in Khovanov homology is [KWZ19, Thm. 8.1].

The pretzel case. From now on $q \geq 7$ is odd and $m = (q - 5)/2$, so that

$$S_{-m}(\gamma, \theta) = (\gamma, \theta + (q - 5)\gamma).$$

By Theorem 4.4 and Lemma 4.6, the decomposition $K_q = (Y, T) \cup_{t_e^{-m}} (D, U)$ has restriction map $S_{-m} \circ L_1$, where L_1 is the restriction map of [Wue26c] for $T(3, 5)$. On $\mathbb{R}^2$ we keep $\varphi = \theta - \gamma$; then

$$\varphi \circ S_{-m} = \lambda, \quad \lambda(\gamma, \theta) = \theta + (q - 6)\gamma,$$

and $S_{-m}^{-1} \ell_1$ is the line field spanned by $(1, 6 - q)$.

4.3. The unperturbed curve. We use the following facts about the tangle (Y, T) for $n = 5$ and its holonomy perturbations, in the notation of [Wue26c, §§3–6] with these changes: the perturbation parameters, written $\epsilon = (\epsilon_A, \epsilon_B)$ there, are written $\varepsilon = (\varepsilon_A, \varepsilon_B)$ here; the binary dihedral arc B of [Wue26c], its pieces B_0, B_1, B_2 and the vectors T_B, P_B are written $\Omega, \Omega_0, \Omega_1, \Omega_2, T_\Omega, P_\Omega$, to distinguish them from the generator B ; the lines Δ_k and the strands $S_\pm^k$ of [Wue26c] are the lines $\mathcal{D}_k$ and the lifts $S_\pm^k$ of L_0 of §2.2; and the reduced function g of [Wue26c, §5.2] is written f_c .

(P1) *The perturbation* [Wue26c, §3.1]. The group $\pi_1(Y \setminus T)$ is free on A, B , and the holonomy perturbation π is supported near two curves P_A, P_B in Y , those of [HHK18, §10], with perturbation functions $\varepsilon_A \sin$ and $\varepsilon_B \sin$. A perturbed traceless representation is determined, up to conjugation, by

$$A \mapsto e^{uQ_A}, \quad B \mapsto e^{vQ_B}, \quad Q_A = \mathbf{i}, \quad Q_B = \tau \mathbf{i} + \sqrt{1 - \tau^2} \mathbf{j},$$

with $(u, v, \tau) \in [0, \pi]^2 \times [-1, 1]$.

(P2) *The equations* [Wue26c, §3.1 and Lemma 3.1], [HHK18, (33)–(34)]. The meridians a, b of T are sent to traceless elements exactly when $\Psi = (\Psi_1, \Psi_2) = 0$, where $\Psi_i = \cos \Theta_i \cos \Phi_i - \tau \sin \Theta_i \sin \Phi_i$ and, for $n = 5$, the perturbed phases are

$$\Phi_1 = 2u + 3\varepsilon_A \sin u, \quad \Theta_1 = 3v - 2\varepsilon_B \sin v, \quad \Phi_2 = -u - 2\varepsilon_A \sin u, \quad \Theta_2 = -2v + \varepsilon_B \sin v.$$

Put $W_\varepsilon = \Psi(\varepsilon; \cdot)^{-1}(0) \subset [0, \pi]^2 \times [-1, 1]$ and $W = W_0$. There is a surjection $W_\varepsilon \rightarrow R_\pi(Y, T)$ [HHK18, Thm. 10.2], whose fiber over the image of (u, v, τ) is that single point unless $\sin u = 0$ or $\sin v = 0$, in which case it is the segment $\{(u, v)\} \times [-1, 1]$. At $\varepsilon = 0$, $\Psi = (E2, E1)$, where (E1) and (E2) denote

$$\cos 2v \cos u - \tau \sin 2v \sin u \quad \text{and} \quad \cos 3v \cos 2u - \tau \sin 3v \sin 2u.$$

The restriction map $\Pi = \Pi_\varepsilon: W_\varepsilon \rightarrow P$ is normalized by $a \mapsto \mathbf{i}$, $b \mapsto e^{\gamma \mathbf{k}} \mathbf{i}$, $c \mapsto e^{\theta \mathbf{k}} \mathbf{i}$ with $\gamma \in [0, \pi]$, and it depends smoothly on $(\varepsilon, u, v, \tau)$ where it is defined [Wue26c, §§3.1, 5].

The unperturbed variety W is described as follows.

- (V1) $W = \Omega \cup C_0$, where $\Omega = \{(\frac{\pi}{2}, \frac{\pi}{2})\} \times [-1, 1]$ is the binary dihedral arc and C_0 is the central circle; $C_0 \cap \Omega = \{c_+, c_-\}$ with $c_\pm = (\frac{\pi}{2}, \frac{\pi}{2}, \pm \tau_1)$, $\tau_1 = \sqrt{3}/2$; the halves $C_0^L = C_0 \cap \{v \leq \frac{\pi}{2}\}$ and $C_0^R = C_0 \cap \{v \geq \frac{\pi}{2}\}$ meet $\{v = \frac{\pi}{2}\}$ only at $c_\pm$ [Wue26c, Prop. 3.6, Prop. 3.10(2)].
- (V2) $\Pi(\Omega)$ lifts to the segment $t \mapsto (t, 0)$, $t \in [0, \pi]$, with $\gamma = \arccos \tau$ on Ω . The endpoints $r_\pm = (\frac{\pi}{2}, \frac{\pi}{2}, \pm 1)$ map to $(0, 0)$ and $(\pi, 0)$, and $\Pi(c_+) = (\frac{\pi}{6}, 0)$, $\Pi(c_-) = (\frac{5\pi}{6}, 0)$ [Wue26c, Prop. 3.9]. We write $\Omega_0, \Omega_1, \Omega_2$ for the pieces of Ω with $\tau \in [\tau_1, 1]$, $[-\tau_1, \tau_1]$, $[-1, -\tau_1]$.
- (V3) Along each half of C_0 , from c_+ to c_- , φ is strictly monotone with $d\varphi \neq 0$ at interior points and increases by $10\pi/3$; at interior points $\frac{\pi}{6} < \gamma < \frac{5\pi}{6}$ [Wue26c, Thm. 4.10, Props. 4.11–4.12].
- (V4) v is a regular parameter of C_0 at $c_\pm$, and on a lift $\Pi = \Pi(c_\pm) \pm V_0(v - \frac{\pi}{2})^2 + O(|v - \frac{\pi}{2}|^3)$ with $V_0 = (2/\sqrt{3}, 12/\sqrt{3})$, of slope 6 [Wue26c, Lemma 4.13].
- (V5) $c_\pm$ are nondegenerate double points of W . With respect to the equations (E1), (E2) of (P2), the kernel of $d\Psi(c_\pm)$ is spanned by $T_\Omega = \partial_\tau$ and $X = -2\tau\partial_u + \partial_v$, the covector $\ell = (-2\tau, -1)$ spans the left kernel, $\ell \cdot D^2\Psi(T_\Omega, X) = -8\tau$ and $\ell \cdot D^2\Psi(T_\Omega, T_\Omega) = 0$ [Wue26c, Prop. 3.11(2)].

Lemma 4.7. *Let $\varepsilon = 0$ and put $w = (ba)^{-3}c(ba)^3a \in \pi_1(Y \setminus T)$. Then $\rho(w) = 1$ for every $\rho \in C_0$. Consequently $\theta \equiv 6\gamma - \pi \pmod{2\pi}$ on $\Pi(C_0)$, and for $l \in \mathbb{Z}$ the lift of each half of C_0 that starts at the point $(\frac{\pi}{6}, 2\pi l)$ over $\Pi(c_+)$ is the segment*

$$\{(\gamma, 6\gamma - \pi + 2\pi l) : \frac{\pi}{6} \leq \gamma \leq \frac{5\pi}{6}\},$$

traversed with γ strictly increasing from c_+ to c_- and with $d\gamma \neq 0$ at interior points.

Proof. We write quaternions as $p = p_0 + \mathbf{p}$ with $p_0 = \operatorname{Re} p$, so that $\mathbf{p}\mathbf{q} = -\mathbf{p} \cdot \mathbf{q} + \mathbf{p} \times \mathbf{q}$ for pure quaternions.

Step 1: the word. By (4.1), $ba = B^{-2}(AB)B^2$, so $(ba)^3 = B^{-2}w_0$ with $w_0 = (AB)^3B^2$. Since $c = B^{-2}aB^2$, we get $w = w_0^{-1}B^2 \cdot B^{-2}aB^2 \cdot B^{-2}w_0 \cdot a = w_0^{-1}aw_0a$. If $\rho(a)$ and $\rho(w_0)$ are orthogonal pure unit quaternions, they anticommute, so $\rho(w_0)^{-1}\rho(a)\rho(w_0) = -\rho(a)$ and $\rho(w) = -\rho(a)^2 = 1$. It remains to prove that orthogonality.

Step 2: notation. Let $\rho \in W$. Then $\beta = \rho(b)$ is a pure unit quaternion. Write $N = \rho(B) = \cos v + \sin v Q$ with $Q = Q_B$, and put

$$\zeta = \beta \cdot Q, \quad U = \cos v \beta + \sin v (\beta \times Q), \quad E = 4 \cos^2 v + 4 \zeta^2 \sin^2 v - 3.$$

From $b = B^{-2}A^{-1}$ we get $\rho(A) = -\beta N^{-2}$.

Step 3: $\rho(a)$. Conjugation by β is the rotation by π about β , so $\beta Q \beta^{-1} = Q'$ with $Q' = 2\zeta\beta - Q$, and $\beta N^{-2}\beta = -e^{-2vQ'}$. Hence $\rho(a) = \rho(A)^2 N^3 = -e^{-2vQ'} N$. Using $\operatorname{Re}(e^{sP} e^{tR}) = \cos s \cos t - (P \cdot R) \sin s \sin t$ and $Q' \cdot Q = 2\zeta^2 - 1$,

$$\operatorname{Re} \rho(a) = -\cos v \cdot E, \quad \mathbf{Vec} \rho(a) = -\sin 3v Q + 2\zeta \sin 2v U.$$

Step 4: $C_0 \subset \{E = 0\}$. On W the element $\rho(a)$ is traceless, so $\cos v \cdot E = 0$. On $C_0 \setminus \{c_{\pm}\}$ we have $v \neq \frac{\pi}{2}$ by (V1), so $E = 0$ there, and by continuity on all of C_0 .

Step 5: $\rho(w_0)$. Put $\xi = \beta N^{-1}$, so that $\rho(AB) = -\xi$ and $\operatorname{Re} \xi = \zeta \sin v$. The Cayley–Hamilton identity $\xi^2 = 2(\operatorname{Re} \xi)\xi - 1$ gives $\xi^3 = (4\zeta^2 \sin^2 v - 1)\xi - 2\zeta \sin v$, and $\xi N^2 = \beta N$, so

$$\rho(w_0) = -\xi^3 N^2 = -(4\zeta^2 \sin^2 v - 1)\beta N + 2\zeta \sin v N^2, \quad \operatorname{Re} \rho(w_0) = \zeta \sin v \cdot E.$$

On $\{E = 0\}$ we have $4\zeta^2 \sin^2 v - 1 = -2 \cos 2v$, so $\rho(w_0)$ is pure, with $\mathbf{Vec} \rho(w_0) = 2 \cos 2v U + 2\zeta \sin v \sin 2v Q$.

Step 6: orthogonality. On $\{E = 0\}$, with $\mathbf{Vec} \rho(w_0)$ in the form of Step 5, and with $U \cdot U = 1 - \zeta^2 \sin^2 v$, $U \cdot Q = \zeta \cos v$ and $\sin 3v = \sin v (4 \cos^2 v - 1)$, a direct expansion gives

$$\mathbf{Vec} \rho(w_0) \cdot \mathbf{Vec} \rho(a) = 2\zeta (2\zeta^2 \sin^2 v \sin 2v + 2 \cos 2v \sin 2v - \sin 3v \cos v) = \zeta \sin 2v \cdot E.$$

So on C_0 the quaternions $\rho(a)$ and $\rho(w_0)$ are orthogonal pure units, and $\rho(w) = 1$ by Step 1.

Step 7: the pillowcase. In the normalization of Π , $\rho(ba) = -e^{\gamma \mathbf{k}}$ and $\mathbf{i} e^{t\mathbf{k}} = e^{-t\mathbf{k}} \mathbf{i}$, so $\rho((ba)^j c (ba)^{-j} a) = -e^{(\theta + 2j\gamma)\mathbf{k}}$. For $j = -3$, $\rho(w) = 1$ gives $\theta - 6\gamma \in \pi + 2\pi\mathbb{Z}$.

Step 8: the lifts. By (V3) the lift of a closed half starting at a lift $(\frac{\pi}{6}, 2\pi l)$ of $\Pi(c_+)$ stays in the open strip except at its ends. On it $\theta - 6\gamma + \pi$ is continuous with values in $2\pi\mathbb{Z}$, hence equal to its value $2\pi l$ at the start, so $\varphi = 5\gamma - \pi + 2\pi l$. By (V3), φ is strictly monotone with $d\varphi \neq 0$ at interior points and increases by $10\pi/3$; hence $\gamma = (\varphi + \pi - 2\pi l)/5$ increases strictly from $\frac{\pi}{6}$ to $\frac{5\pi}{6}$, with $d\gamma \neq 0$ at interior points. $\square$

Each displayed identity of Steps 1–7 has also been verified symbolically for indeterminate (v, ζ) (see Appendix A). Hedden, Herald and Kirk observed, by a computer algebra computation, that the two halves of the central circle for $T(3, 5)$ map linearly with slope 6 [HHK18, §11.7, p. 59]; Lemma 4.7 proves this. For their own decomposition of the pretzel knots $P(p, q, r)$ with p even, Fukumoto, Kirk and Pinzón-Caicedo prove the analogous straightness [FKP17, Thm. 5.1], and they describe the unperturbed curve as the segments $\theta = (r - 5)\gamma$ and $\theta = (r + 1)\gamma + \pi$ [FKP17, §5, p. 28]. These are the images under S_{-m} of $\Pi(\Omega)$ and of the line of Lemma 4.7, with $r = q$. We do not identify their decomposition with ours, and we do not use their result; we regard it as the antecedent of Lemma 4.7.

By (P6) in §4.4, after a small perturbation the arc of L_1 is built from Ω_0 , one half of C_0 and Ω_2 , possibly through Ω_1 and the other half. Write H for the half joined to Ω_0 and H' for the other half, and orient both from c_+ to c_- . Let $O = (0, 0)$, and

$$J_1 = (\frac{\pi}{6}, 0), \quad J_2 = (\frac{5\pi}{6}, 4\pi), \quad J_3 = (\frac{\pi}{6}, 4\pi), \quad J_4 = (\frac{5\pi}{6}, 8\pi), \quad Z = (\pi, 8\pi), \quad Z' = (\pi, 4\pi).$$

Corollary 4.8. *At $\varepsilon = 0$ the lifted pieces are straight segments, traversed monotonically and regular in their interiors: $\Omega_0: O \rightarrow J_1$ along $(1, 0)$; $H: J_1 \rightarrow J_2$ along $(1, 6)$; $\Omega_1^{-1}: J_2 \rightarrow J_3$ along $(-1, 0)$; $H': J_3 \rightarrow J_4$ along $(1, 6)$; and $\Omega_2: J_4 \rightarrow Z$, or $\Omega_2: J_2 \rightarrow Z'$ when Ω_2 follows H , along $(1, 0)$. The lift of the closed curve $\Omega_1^{-1} \cdot H'$ is the $(0, 4\pi)$ -periodic polygon with vertices $(\frac{5\pi}{6}, 4\pi j)$ and $(\frac{\pi}{6}, 4\pi j)$, $j \in \mathbb{Z}$.*

Proof. The pieces of Ω are described by (V2): $\gamma = \arccos \tau$ is a diffeomorphism from $(-1, 1)$ onto $(0, \pi)$. The halves are described by Lemma 4.7, applied to the lifts starting at $J_1 = (\frac{\pi}{6}, 0)$ and $J_3 = (\frac{\pi}{6}, 4\pi)$, whose endpoints are the listed vertices. $\square$

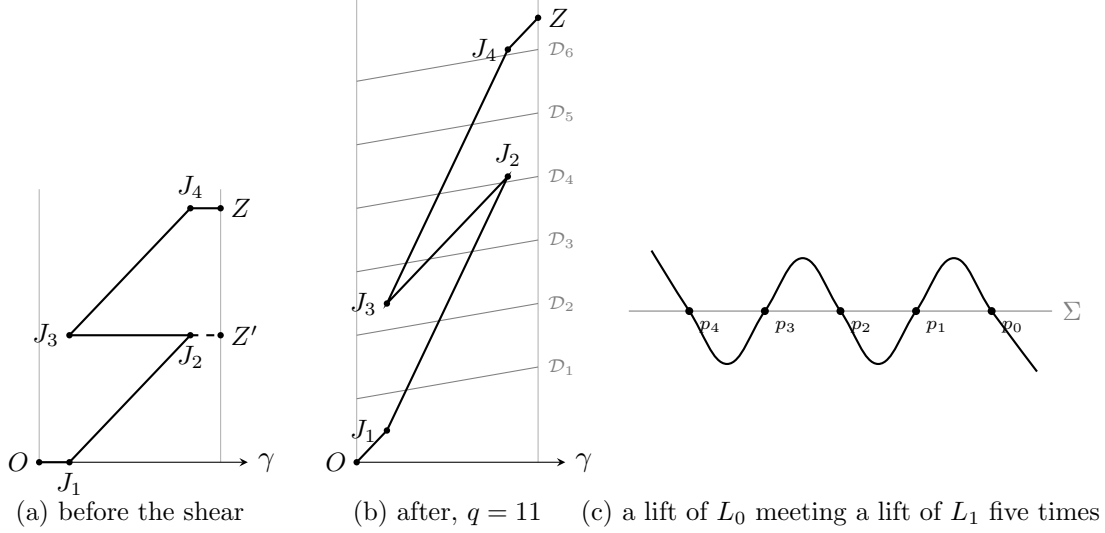


FIGURE 1. (a) The lift of the unperturbed curve for $T(3, 5)$, at $\varepsilon = 0$ (Corollary 4.8); the dashed segment is Ω_2 when it follows H . Vertical units are multiples of π . (b) Its image under S_{-m} for $q = 11$, with the lines $\mathcal{D}_k$; the lines $\mathcal{D}_3$ and $\mathcal{D}_4$ are crossed three times. (c) The configuration of Lemma 4.16: the points $p_0, \dots, p_4$ are ordered by decreasing γ , and the lobes alternate above and below the lift Σ of L_0 .

4.4. The perturbed curves. Besides (P1) and (P2) we use the following results of [Wue26c, §§5–6], stated for $n = 5$. In them c is one of the double points $c_{\pm}$ of (V1), ℓ is the covector of (V5), and L_1 and $\tilde{A}$ are the curve of [Wue26c] and its lift, before the shear; statements that involve L_0 hold, as in [Wue26c, §6], for $\varepsilon \in \mathcal{C}_a$ sufficiently small and then every sufficiently small earring parameter.

(P3) *The reduced function* [Wue26c, §5.2]. The differential $d\Psi(c)$ has rank one, its kernel is spanned by T_{Ω} and the tangent T_C of C_0 at c , and the restriction of $\ell \cdot D^2\Psi(c)$ to $\ker d\Psi(c)$ is a nondegenerate indefinite quadratic form [Wue26c, Prop. 3.11]. In coordinates (s, t, ϱ) centered at c with $\partial_s = T_{\Omega}$ and $\partial_t = T_C$ at c , for a covector m not proportional to ℓ the function $m \cdot \Psi$ has nonzero ∂_{ϱ} -derivative at c and can be solved for $\varrho = \varrho(s, t, \varepsilon)$ by the implicit function theorem, and near c the variety W_{ε} is the zero set of the *reduced function*

$$f_c(s, t; \varepsilon) = \ell \cdot \Psi(\varepsilon; s, t, \varrho(s, t, \varepsilon)),$$

that is, of the Lyapunov–Schmidt reduction of Ψ to $\ker d\Psi(c)$; its Hessian at $(0, 0; 0)$ is the quadratic form above. For small ε the function $f_c(\cdot; \varepsilon)$ has a unique critical point near 0, depending analytically on ε , with critical value $\kappa_c(\varepsilon)$, and there are coordinates (s', t') , depending analytically on ε , with $f_c = \kappa_c(\varepsilon) + s't'$; at $\varepsilon = 0$ they are chosen so that $\{t' = 0\}$ is Ω and $\{s' = 0\}$ is C_0 near c . Moreover $\kappa_c(0) = 0$ and $d\kappa_c(0) = \ell \cdot \partial_{\varepsilon}\Psi(0; c)$.

(P4) *The cone* [Wue26c, Def. 5.5]. For $a > 0$, $\mathcal{C}_a$ is the set of $\varepsilon \in \mathbb{R}^2$ such that the linear part $d\kappa_c(0)\varepsilon$ of $\kappa_c(\varepsilon)$ has absolute value at least $a|\varepsilon|$ at both double points $c = c_{\pm}$, for the covector ℓ of (V5); another normalization of ℓ rescales a .

(P5) *Near the double points* [Wue26c, Lemma 5.6 and its proof]. Let P_{Ω} and V be the vectors with $\Pi(s', 0) = \Pi(c) + s'P_{\Omega} + O(s'^2)$ and $\Pi(0, t') = \Pi(c) + t'V + O(t'^3)$ at $\varepsilon = 0$; they are linearly independent. Let $a > 0$. There are $r_1 > 0$ and constants $C, c_0 > 0$ such that for every $r_0 \in (0, r_1]$ there is $\varepsilon_0 > 0$ with the following properties for every $\varepsilon \in \mathcal{C}_a$ with

$0 < |\varepsilon| < \varepsilon_0$. Let U_c be the set of points of a fixed neighborhood of c whose coordinates (s, t) have Morse coordinates (s', t') with $|s'| \leq r_0$ and $|t'| \leq r_0$; it depends on ε , but it contains and is contained in fixed neighborhoods of c . Each of the two arcs of $W_\varepsilon \cap U_c$ is mapped by Π to an embedded arc with nowhere vanishing velocity. Such an arc is parametrized by t' , with $s' = -\kappa/t'$, $\kappa = \kappa_c(\varepsilon)$ and $|\kappa|/r_0 \leq |t'| \leq r_0$, and its velocity is

$$\frac{d}{dt'} \Pi_\varepsilon = \frac{\kappa}{t'^2} P_\Omega + 2t'V + e(t'), \quad |e(t')| \leq C(r_0 + |\varepsilon|^{2/3} a^{-1/3}) A(t'), \quad A(t') = \frac{|\kappa|}{t'^2} + |t'|,$$

with C independent of ε , a and r_0 ; the main term $(\kappa/t'^2)P_\Omega + 2t'V$ has length at least $c_0 A(t')$.

- (P6) *Global structure* [Wue26c, Prop. 5.7(1), (2), (4)]. Let $a > 0$. For every $\varepsilon \in \mathcal{C}_a$ with $0 < |\varepsilon|$ sufficiently small, $R_\pi(Y, T)$ is a compact 1-manifold with two boundary points $r_\pm(\varepsilon)$, and L_1 is an immersion into P^* away from $r_\pm(\varepsilon)$. Away from $U_{c_\pm}$ and from the corner charts of (P7), W_ε is C^1 -close to W : it is the image of a C^1 -small normal graph over W , with Π_ε correspondingly C^1 -close to Π . The union $\Omega \cup C_0$ is replaced by its resolution at the two double points: either the arc is $\Omega_0 \cdot C_0^X \cdot \Omega_2$ and there is one further circle $\Omega_1 \cdot (C_0^Y)^{-1}$, or the arc is $\Omega_0 \cdot C_0^X \cdot \Omega_1^{-1} \cdot C_0^Y \cdot \Omega_2$ and there is no further circle, where $\{X, Y\} = \{L, R\}$ and each product denotes the resolved curve, which is C^1 -close to the listed pieces away from $c_\pm$ and is described by (P5) near them. For $n = 5$ the variety W has no non-central circles [Wue26c, Prop. 3.6].
- (P7) *Near the corners* [Wue26c, Lemma 5.1 and Cor. 5.3]. There are a neighborhood U_r of $r_\pm(0)$, a number $y_0 > 0$ and a neighborhood $\mathcal{O}$ of $0 \in \mathbb{R}^2$ such that for every $\varepsilon \in \mathcal{O}$, $R_\pi(Y, T) \cap U_r$ is a single embedded half-open arc $\{\Gamma_\varepsilon(y) : 0 \leq y < y_0\}$ with $\Gamma_\varepsilon(0) = r_\pm(\varepsilon)$, Π maps $r_\pm(\varepsilon)$ to the corner $\Pi(r_\pm(0))$, and, with a suitable lift, $\tilde{\Pi}(\Gamma_\varepsilon(y)) = y\mathbf{c}(\varepsilon) + y^3 R(y, \varepsilon)$, where $\mathbf{c}(\varepsilon) = (c_1(\varepsilon), c_2(\varepsilon)) \in \mathbb{R}^2$ and R are real-analytic and $\mathbf{c}(0)$ is a nonzero vector along the image of Ω . With constants independent of ε : the limiting slope of L_1 at r_+ is the slope of the image of Ω plus $O(|\varepsilon|)$; on the lift through the lattice point, $\varphi = y(c_2(\varepsilon) - c_1(\varepsilon)) + O(y^3)$ with $c_2 - c_1$ bounded away from 0, so that φ is strictly monotone on $0 < y < y_0$; and if U_r is the neighborhood of $r_+(0)$, then for every sufficiently small earring parameter L_0 meets $L_1(R_\pi(Y, T) \cap U_r)$ in exactly one point, the generator r_+ of [HHK18, Lemma 5.1]. The statements about the limiting slope and about φ also hold at the other end of the arc, whose image is the corner $(\pi, 0)$.
- (P8) *The arc and the remnant circle* [Wue26c, §6.1, Lemmas 6.1 and 6.2]. Let $\tilde{A}$ be the lift of the arc component of L_1 that starts at $(0, 0)$ and enters $\{\gamma > 0\}$. If the arc is $\Omega_0 \cdot H \cdot \Omega_1^{-1} \cdot H' \cdot \Omega_2$, then $\tilde{A}$ joins $(0, 0)$ to $(\pi, 8\pi)$, and if it is $\Omega_0 \cdot H \cdot \Omega_2$, to $(\pi, 4\pi)$. The interior of $\tilde{A}$ lies in the open strip $0 < \gamma < \pi$, $\tilde{A}$ is embedded, and φ along $\tilde{A}$ is piecewise monotone, taking the values $0, -\frac{\pi}{6}, 3\pi + \frac{\pi}{6}, 4\pi - \frac{\pi}{6}, 7\pi + \frac{\pi}{6}, 7\pi$, resp. $0, -\frac{\pi}{6}, 3\pi + \frac{\pi}{6}, 3\pi$, at the ends of the pieces of the unperturbed curve, up to $O(|\varepsilon|)$. When the arc is $\Omega_0 \cdot H \cdot \Omega_2$, the further circle of (P6), the *remnant circle* $\Omega_1^{-1} \cdot H'$, misses the left and right edges, φ is strictly monotone along it with $d\varphi \neq 0$, and φ changes by 4π over one period; its vertical degree is 2.
- (P9) *Bigons and admissibility* [Wue26c, Lemmas 6.4 and 6.6]. The boundary of a bigon from p to p' lifts to a closed curve in $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2$ consisting of a sub-path $\tilde{\beta}$ of one lift of a component of L_1 , from a lift $\tilde{p}'$ of p' to a lift $\tilde{p}$ of p , and a sub-arc of one lift $S_\pm^k$ of L_0 from $\tilde{p}$ to $\tilde{p}'$. If φ is strictly monotone along $\tilde{\beta}$, then no such bigon exists; in particular a component of L_1 along which φ is strictly monotone carries no bigon. Moreover L_1 is a restricted immersed 1-manifold, and (L_0, L_1) is an admissible pair.

By (V2) and (V4), $P_\Omega \parallel (1, 0)$ and $V \parallel \pm(1, 6)$. The first statement of (P9) holds with any immersed curve in place of L_1 , because $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2 \rightarrow P^*$ is a covering. Where we apply an argument of [Wue26c] to the sheared curve, rather than a statement of (P5)–(P9) to the curve of [Wue26c], we say so.

For $\varepsilon \in \mathcal{C}_a$ small, each of $c_\pm$ is resolved: the two arcs of W_ε near c each join a half-branch of Ω to a half-branch of C_0 . Following [Wue26c, §6.1], we name the two alternatives of (P6):

- the *joined resolution*, in which $R_\pi(Y, T)$ is a single arc $\Omega_0 \cdot H \cdot \Omega_1^{-1} \cdot H' \cdot \Omega_2$;
- the *split resolution*, in which $R_\pi(Y, T)$ is the arc $\Omega_0 \cdot H \cdot \Omega_2$ together with the remnant circle $\Omega_1^{-1} \cdot H'$.

Here each product denotes the resolved curve, as in (P6). The analysis of [Wue26c] does not need to know which type occurs. After the shear it matters, and we determine it.

Lemma 4.9. *Let $c \in \{c_+, c_-\}$, $a > 0$, and let λ' be a linear function on $\mathbb{R}^2$ with $\lambda'(P_\Omega) \neq 0$ and $\lambda'(V) \neq 0$. Let U_c (the box of half-width r_0 in the Morse coordinates) and ε_0 be as in (P5). After shrinking r_0 and ε_0 , the following holds for every $\varepsilon \in \mathcal{C}_a$ with $0 < |\varepsilon| < \varepsilon_0$ and each of the two arcs of $W_\varepsilon \cap U_c$, traversed from its end on Ω to its end on C_0 or conversely.*

- (1) *Call arrival direction the direction of traversal along the first half-branch towards c , and departure direction that along the second half-branch away from c ; each is a multiple of $\pm P_\Omega$ or $\pm V$. For any prescribed $\vartheta > 0$, the velocity of Π along the arc stays within angle ϑ of the closed convex cone spanned by the arrival and departure directions, which has opening less than π . Consequently, if ℓ' is a constant line field not parallel to the arrival or departure direction, and ϑ is small, the net number of passages of the tangent line through ℓ' is ± 1 if one of the two directions of ℓ' lies inside that cone, the sign being that of the rotation from the arrival to the departure direction, and 0 otherwise.*
- (2) *If the two joined half-branches move λ' in opposite senses away from c , then $\lambda' \circ \Pi$ is strictly monotone along the arc.*
- (3) *Along the arc, $|\lambda' \circ \Pi - \lambda'(\Pi(c))| \leq C_{\lambda'}(r_0 + |\varepsilon|)$, with $C_{\lambda'}$ independent of ε .*

Proof. By (P5) the arc is parametrized by t' , with $s' = -\kappa/t'$, $\kappa = \kappa_c(\varepsilon)$ and $|\kappa|/r_0 \leq |t'| \leq r_0$, and its velocity is $(\kappa/t'^2)P_\Omega + 2t'V + e(t')$ with $|e| \leq C(r_0 + |\varepsilon|^{2/3}a^{-1/3})A(t')$, where $A(t') = |\kappa|/t'^2 + |t'|$ and C does not depend on r_0 . Along the arc the signs of κ and t' are fixed, so the main term is a positive combination of $\text{sign}(\kappa)P_\Omega$ and $\text{sign}(t')V$, of length at least $c_0A(t')$. At the end where $|t'| \approx r_0$ the term $2t'V$ dominates, and at the end where $|t'| \approx |\kappa|/r_0$ the term $(\kappa/t'^2)P_\Omega$ does; these are the directions of traversal along the two half-branches. Shrinking r_0 and ε_0 makes $|e|/A$ as small as we wish, which gives the first assertion of (1). The tangent direction moves continuously in a sector of opening less than π , which gives the count of passages. For (2), λ' applied to the main term is $\kappa\lambda'(P_\Omega)/t'^2 + 2\lambda'(V)t'$. In the case of opposite senses both summands have the same sign, so its absolute value is at least a fixed multiple of A , and the error does not change the sign. For (3), Π_ε is uniformly Lipschitz on U_c and $\Pi_\varepsilon(c) = \Pi(c) + O(|\varepsilon|)$, by the smooth dependence in (P2). $\square$

For $\lambda' = \lambda$ we have $\lambda(1, 0) = q - 6 \neq 0$ and $\lambda(1, 6) = q \neq 0$, so the lemma applies at c_+ and c_- .

Proposition 4.10. *Let $a > 0$ and $\varepsilon \in \mathcal{C}_a$ with $|\varepsilon|$ small. The critical values at $c_\pm$ are*

$$\kappa_\pm(\varepsilon) = \mp \frac{\sqrt{3}}{2} \varepsilon_A - \frac{1}{2} \varepsilon_B + O(|\varepsilon|^2),$$

for the covector $\ell = (-2\tau, -1)$ of (V5). The resolution is

$$\text{joined} \iff |\varepsilon_B| < \sqrt{3}|\varepsilon_A|, \quad \text{split} \iff |\varepsilon_B| > \sqrt{3}|\varepsilon_A|.$$

For small a both types occur in $\mathcal{C}_a$: the joined sectors contain the ε_A -axis, and the split sectors the ε_B -axis. The set excluded from $\mathcal{C}_a$ is an open conical neighborhood of the two lines $\{\varepsilon_B = \mp\sqrt{3}\varepsilon_A\}$, the boundaries between the sectors.

Proof. At $c = (\frac{\pi}{2}, \frac{\pi}{2}, \tau)$, $\tau = \pm\tau_1$, put $\Lambda = \ell \cdot D^2\Psi(T_\Omega, X) = -8\tau$ (V5). The tangent T_C of C_0 at c is the second null direction of the reduced Hessian; it has the form $X + \alpha T_\Omega$, with ∂_v -component 1 because v is a regular parameter of C_0 at c (V4). Since $\ell \cdot D^2\Psi(T_\Omega, T_\Omega) = 0$, we have $\ell \cdot D^2\Psi(T_\Omega, T_C) = \Lambda$. So in coordinates (s, t) along (T_Ω, T_C) the reduced function of (P3) is $f_c = \kappa_c(\varepsilon) + \Lambda st + (\text{higher order})$, and the arc of $\{f_c = 0\}$ in the quadrant with $\text{sign}(s)\text{sign}(t) = -\text{sign}(\kappa_c\Lambda)$ joins the half-branch of Ω in direction $\text{sign}(s)T_\Omega$ to the half-branch of C_0 in direction $\text{sign}(t)T_C$. Now $+T_C$ points into C_0^R at both double points, and $+T_\Omega = \partial_\tau$ points into Ω_0 at c_+ and into Ω_1 at c_- . At c_+ , $\Lambda < 0$, so Ω_0 is joined to the half with $\text{sign}(t) = \text{sign}(\kappa_+)$; at c_- , $\Lambda > 0$, so Ω_2 (where $s < 0$) is joined to the half with $\text{sign}(t) = \text{sign}(\kappa_-)$. The resolution is split exactly when Ω_0 and Ω_2 are joined to the same half, that is, when $\kappa_+\kappa_- > 0$.

For the critical values we use the phases Φ_i, Θ_i of (P2); since $\Psi = (\text{E2}, \text{E1})$ at $\varepsilon = 0$, the covector ℓ of (V5) pairs with Ψ as $\ell \cdot \Psi = -2\tau\Psi_2 - \Psi_1$. At c the phases give $\partial_\varepsilon\Psi_1 = (-3\tau, 2)$ and $\partial_\varepsilon\Psi_2 = (2, -\tau)$, so $d\kappa_c(0) = \ell \cdot \partial_\varepsilon\Psi = -2\tau(2, -\tau) - (-3\tau, 2) = (-\tau, 2\tau^2 - 2) = (-\tau, -\frac{1}{2})$. Hence $\kappa_+\kappa_- = \frac{1}{4}\varepsilon_B^2 - \frac{3}{4}\varepsilon_A^2 + O(|\varepsilon|^3)$. On $\mathcal{C}_a$ both linear parts are at least $a|\varepsilon|$ in absolute value, so for $|\varepsilon|$ small the signs of $\kappa_\pm$ are those of their linear parts. $\square$

In particular the proposition decides which alternative of (P6) occurs for $n = 5$.

Let $\tilde{A}$ be the lift of the arc of (P8), starting at O , and $\hat{A} = S_{-m}(\tilde{A})$. In the split case let $\tilde{C}$ be the lift of the remnant circle through the lifts of Corollary 4.8, and $\hat{C} = S_{-m}(\tilde{C})$. The values of φ at the sheared vertices are

$$\begin{aligned} \varphi(S_{-m}O) &= 0, & \varphi(S_{-m}J_1) &= \frac{(q-6)\pi}{6}, & \varphi(S_{-m}J_2) &= 4\pi + \frac{5(q-6)\pi}{6}, & \varphi(S_{-m}J_3) &= 4\pi + \frac{(q-6)\pi}{6}, \\ \varphi(S_{-m}J_4) &= 8\pi + \frac{5(q-6)\pi}{6}, & \varphi(S_{-m}Z) &= (q+2)\pi, & \varphi(S_{-m}Z') &= (q-2)\pi. \end{aligned} \quad (4.5)$$

We write $\varphi_i = \varphi(S_{-m}J_i)$. Since $q-6$ is odd, each φ_i differs from $4\pi\mathbb{Z}$ by an odd multiple of $\pi/6$, and so lies at distance at least $\pi/6$ from $2\pi\mathbb{Z}$.

Proposition 4.11. *Let $a > 0$. There are neighborhoods U_{J_i} of the junctions $S_{-m}J_i$ (the images of $U_{c_\pm}$) and of the corners such that for every $\varepsilon \in \mathcal{C}_a$ with $0 < |\varepsilon|$ sufficiently small, the threshold depending on q :*

- (1) $\hat{A}$ is an embedded proper arc from $(0, 0)$ to $(\pi, (q+3)\pi)$ in the joined case and to $(\pi, (q-1)\pi)$ in the split case, with interior in the open strip. It leaves $(0, 0)$ with limiting slope $q-5 + O(|\varepsilon|)$, and $\varphi > 0$ on its initial segment.
- (2) Outside $\bigcup_i U_{J_i}$ and the corner neighborhoods, $\hat{A}$ is C^1 -close to the image under S_{-m} of the polygon of Corollary 4.8, and φ is strictly monotone with $|d\varphi|$ bounded below: increasing near $S_{-m}(\Omega_0)$, $S_{-m}(H)$, $S_{-m}(H')$, $S_{-m}(\Omega_2)$, and decreasing near $S_{-m}(\Omega_1^{-1})$. In U_{J_1} , in U_{J_4} , and at the junction of H with Ω_2 in the split case, φ is strictly monotone. In U_{J_2} and U_{J_3} , $|\varphi - \varphi_i| < \pi/12$. Near the corners $|d\varphi|$ is bounded below.
- (3) For a sub-arc α of $\hat{A}$ with endpoints outside $\bigcup_i U_{J_i}$, $\mu(\alpha, \ell_1)$ is the number of neighborhoods U_{J_2} traversed by α minus the number of U_{J_3} traversed. All critical points of φ on $\hat{A}$ lie in $U_{J_2} \cup U_{J_3}$.

In the split case $\hat{C}$ is a restricted immersed circle that misses the edges and has vertical degree 2. Its lift is $(0, 4\pi)$ -periodic and alternates between pieces C^1 -close to $S_{-m}(\Omega_1^{-1}) + (0, 4\pi j)$, on which φ decreases from $\varphi_2 + 4\pi j$ to $\varphi_3 + 4\pi j$, and pieces C^1 -close to $S_{-m}(H') + (0, 4\pi j)$, on

which φ increases from $\varphi_3 + 4\pi j$ to $\varphi_2 + 4\pi(j+1)$; the junctions are of the types J_2 and J_3 , and (2), (3) hold on it verbatim.

Proof. (1) By (P8), $\tilde{A}$ is embedded with interior in the strip and ends at Z or Z' , and S_{-m} preserves γ and maps $(\pi, 2\pi j)$ to $(\pi, 2\pi j + (q-5)\pi)$. The limiting slope is that of $\Pi(\Omega)$, which is 0, plus $q-5$, up to $O(|\varepsilon|)$, by (P7) and Lemma 4.6(c). In the corner chart of (P7) the lift is $y(c_1(\varepsilon), c_2(\varepsilon)) + O(y^3)$ with $(c_1, c_2)(0)$ a positive multiple of $(1, 0)$, so on $\hat{A}$ we get $\varphi = y(c_2 + (q-6)c_1) + O(y^3)$, which is positive for small $y > 0$.

(2) Away from the junctions and corners, $\tilde{A}$ is C^1 -close to the polygon of Corollary 4.8, by (P6). The image under S_{-m} of the polygon has velocities that are positive multiples of $(1, q-5)$ on $S_{-m}(\Omega_0)$ and $S_{-m}(\Omega_2)$, of $(1, q+1)$ on $S_{-m}(H)$ and $S_{-m}(H')$, and of $(-1, 5-q)$ on $S_{-m}(\Omega_1^{-1})$, along which $d\varphi$ is $q-6$, q and $-(q-6)$. At the junctions we apply Lemma 4.9 with $\lambda' = \lambda$:

junction	first half-branch, away from c	second half-branch	conclusion
J_1 at c_+ , $\Omega_0 \rightarrow H$	Ω_0 decreases λ	H increases λ	(2): monotone
J_2 at c_- , $H \rightarrow \Omega_1^{-1}$	H decreases λ	Ω_1^{-1} decreases λ	(3)
J_3 at c_+ , $\Omega_1^{-1} \rightarrow H'$	Ω_1 increases λ	H' increases λ	(3)
J_4 at c_- , $H' \rightarrow \Omega_2$	H' decreases λ	Ω_2 increases λ	(2): monotone
split, at c_- , $H \rightarrow \Omega_2$	H decreases λ	Ω_2 increases λ	(2): monotone

Choose r_0 with $C_\lambda r_0 < \pi/24$ and then $|\varepsilon|$ with $C_\lambda |\varepsilon| < \pi/24$. Near the corners we apply the argument of (P7) to the sheared curve, using Lemma 4.6(c): it uses only that the limiting slope differs from 1, and here it is $q-5 \geq 2$.

(3) By Lemma 4.6(f), $\mu(S_{-m} \circ \tilde{\alpha}, \ell_1) = \mu(\tilde{\alpha}, \ell')$, where ℓ' is spanned by $(1, 6-q)$, at angle $-\arctan(q-6)$, equivalently $\pi - \arctan(q-6) \in (\frac{\pi}{2}, \frac{3\pi}{4}]$. We compute on the unsheared curve. On the C^1 -close pieces the directions are near $(\pm 1, 0)$ or $(1, 6)$, and near the corners the slope is $O(|\varepsilon|)$; none of these is parallel to $(1, 6-q)$, so there are no passages there. In U_{J_2} the velocity moves, by Lemma 4.9(1), from near $(1, 6)$ to near $(-1, 0)$ inside the cone of angles $[\arctan 6, \pi]$. This cone contains the direction at angle $\pi - \arctan(q-6)$ and not its opposite, and the rotation is counterclockwise; so the count is $+1$. In U_{J_3} the motion is reversed and the count is -1 . In U_{J_1} , U_{J_4} and the split junction the cone is $[0, \arctan 6]$, which contains neither direction. A critical point of φ on $\tilde{A}$ is a point where the tangent of $\tilde{A}$ is parallel to $(1, 1)$, that is, where the tangent of $\tilde{A}$ is parallel to $(1, 6-q)$; by the above this happens only in $U_{J_2} \cup U_{J_3}$.

For the remnant circle, the properties of being restricted, avoiding the edges and having vertical degree 2 follow from (P8), (P9) and Lemma 4.6(b)–(d). Its pieces are those of Corollary 4.8, with junctions of type J_2 (from H' to Ω_1^{-1} , at c_-) and J_3 (from Ω_1^{-1} to H' , at c_+), and the estimates above apply to it by compactness and periodicity. $\square$

Remark 4.12. The smallness threshold for ε in Proposition 4.11 depends on q through the constant C_λ , which grows linearly in q . The margins that matter in the next subsection are the distances from φ_3 to the nearest value in $2\pi\mathbb{Z}$ above it and from φ_2 to the nearest value in $2\pi\mathbb{Z}$ below it. Both are at least $\pi/6$; the first equals $\pi/6$ exactly when $q \equiv 5 \pmod{12}$, and the second exactly when $q \equiv 11 \pmod{12}$. The order of the crossings in Proposition 4.17 also requires the perturbation to move the crossings by less than half the gaps between their abscissae at $\varepsilon = 0$, which are at least $\pi/(q(q-6))$ and so shrink quadratically in q . Above the threshold a canceling pair of generators can be missing (a line $\mathcal{D}_k$ crossed once instead of three times); in our computations (Remark 4.21) the homology is the same.

4.5. Generators. Put

$$N_q = \#\left\{j \in \mathbb{Z} : \frac{q-6}{12} < j < \frac{5(q-6)}{12}\right\} = \left\lfloor \frac{5(q-6)}{12} \right\rfloor - \left\lfloor \frac{q-6}{12} \right\rfloor;$$

both bounds are non-integers because $q - 6$ is odd. Thus $N_7 = 0$, $N_9 = 1$, $N_{11} = N_{13} = 2$, $N_{15} = 3$ and $N_{17} = N_{19} = 4$. We call a transverse crossing of a lift of L_1 with a line $\mathcal{D}_k$ an *up-crossing* or a *down-crossing* according as φ increases or decreases through $2\pi k$ there.

Proposition 4.13. *For $\varepsilon \in \mathcal{C}_a \setminus \{0\}$ sufficiently small, and then ϵ sufficiently small:*

- (1) L_0 and L_1 meet transversally. The arc carries r_+ , which lies on S_+^0 , and every component carries a pair $x_P^\pm \in S_\pm^k$ over each transverse crossing P of (one period of) its lift with a line $\mathcal{D}_k$, with $k \geq 1$ for the arc. No generator lies over the double point of L_0 .
- (2) In the joined case, $\hat{A}$ crosses $\mathcal{D}_k$ three times (up, down, up) if $\varphi_3 < 2\pi k < \varphi_2$, that is for $k = 2 + j$ with $\frac{q-6}{12} < j < \frac{5(q-6)}{12}$, and once (up) for every other $k \in \{1, \dots, \frac{q+1}{2}\}$; there are no other crossings. The arc carries $q + 2 + 4N_q$ generators.
- (3) In the split case, $\hat{A}$ crosses $\mathcal{D}_1, \dots, \mathcal{D}_{(q-3)/2}$ once each (up) and carries $q - 2$ generators; one period of $\hat{C}$ crosses N_q lines downward and $N_q + 2$ upward, and $\hat{C}$ carries $4 + 4N_q$ generators.

In both cases the complex has $q + 2 + 4N_q$ generators.

Proof. The argument of [Wue26c, Prop. 6.3] applies. By Proposition 4.11(2), outside small neighborhoods of its transverse crossings with the lines $\mathcal{D}_k$ and of the corners, each curve stays a positive distance from $\bigcup_k \mathcal{D}_k$: in U_{J_2} and U_{J_3} the values of φ stay within $\pi/12$ of φ_i , and the same holds in U_{J_1} , U_{J_4} by Lemma 4.9(3). Near a crossing, $|d\varphi|$ is bounded below and the slope is bounded away from 1, so the curve meets each of S_+^k and S_-^k once, transversally. The corner $(0, 0)$ contributes exactly r_+ , by [HHK18, Lemma 5.1] and the argument of (P7) applied to $\hat{A}$. Since $\hat{A}$ leaves $(0, 0)$ with slope $q - 5 > 1$, it meets the lift S_+^0 of L_0 , which passes $(0, 0)$ through $(-\epsilon, \epsilon)$, near $\gamma = 2\epsilon/(q - 6) > 0$, and it does not meet S_-^0 near the corner; so $r_+ \in S_+^0$. The far endpoint lies on $\{\varphi \in (2\mathbb{Z} + 1)\pi\}$, away from every $\mathcal{D}_k$.

In the joined case φ is monotone along the pieces over $(0, \varphi_2)$ (near $S_{-m}(\Omega_0)$ and $S_{-m}(H)$), over (φ_3, φ_2) (near $S_{-m}(\Omega_1^{-1})$) and over $(\varphi_3, (q + 2)\pi)$ (near $S_{-m}(H')$ and $S_{-m}(\Omega_2)$). Here $\varphi_3 > 0$ and $(q + 2)\pi - \varphi_2 = (q + 18)\pi/6 > 0$. The lines with $2\pi k \in (\varphi_3, \varphi_2)$ are those with $k = 2 + j$ and $\frac{q-6}{12} < j < \frac{5(q-6)}{12}$, and the lines with $0 < 2\pi k < (q + 2)\pi$ are $k = 1, \dots, \frac{q+1}{2}$. This gives $1 + 2(\frac{q+1}{2} + 2N_q)$ generators. In the split case φ increases monotonically along the arc over $(0, (q - 2)\pi)$; on one period of the circle it decreases over (φ_3, φ_2) , crossing N_q lines, and then increases over $(\varphi_3, \varphi_2 + 4\pi)$, crossing $N_q + 2$ lines.

At $\varepsilon = 0$ every crossing of the sheared polygons with a line $\mathcal{D}_k$ lies on a piece of Ω , where $\varphi = (q - 6)\gamma + 4\pi j$, or on a half of C_0 , where $\varphi = q\gamma + (2j + 1)\pi$; so its abscissa is $2\pi i/(q - 6)$ or $(2i + 1)\pi/q$ with $i \in \mathbb{Z}$. Neither equals $\pi/2$, because $q - 6$ and q are odd. Hence for small perturbations no crossing lies near the double point $(\frac{\pi}{2}, \frac{\pi}{2})$ of L_0 . $\square$

For $q = 7$ there are $9 = 1 + |\sigma(K_7)|$ generators. In general the count exceeds $1 + |\sigma(K_q)| = q + 2$ by $4N_q$; we compare it with the traceless characters in Remark 4.19.

4.6. Gradings.

Lemma 4.14. *Let p, p' be generators with lifts $\tilde{p} \in S_\pm^k$ and $\tilde{p}' \in S_\pm^{k'}$ on lifts of L_0 of the same sign, joined by a sub-path $\tilde{\alpha}$ of a lift of a component of L_1 that lies in the open strip and has slope bounded away from 1 at $\tilde{p}$ and $\tilde{p}'$. Then*

$$\text{gr}(p, p') \equiv \mu(\alpha_1, \ell_1) + 2(k - k') \pmod{4},$$

where α_1 is the projection of $\tilde{\alpha}$, run from p' to p .

Proof. This is the proof of [Wue26c, Lemma 6.8], without its hypothesis that $\tilde{\alpha}$ contain no critical point of φ . Apply (2.3) with α_0 the branch of L_0 on the front face from p to the point over p' ;

its lift runs along $S_{\pm}^k$ from $\tilde{p}$ to $\tilde{p}' - (0, 2\pi(k' - k))$ inside the open strip. Then $\mu(\alpha_0, \ell_1) = 0$, because the slope of $S_{\pm}^k$ stays on one side of 1. The triple indices at p and p' agree: both are 0 for the sign $+$ and 1 for the sign $-$, since the tangent line of L_1 is bounded away from slope 1. Finally the lift of $\alpha_0 * \alpha_1$ runs in the open strip from $\tilde{p}$ to $\tilde{p} + (0, 2\pi(k - k'))$, so $\alpha_0 * \alpha_1$ is homotopic in P^* to $k - k'$ times the core of the annulus obtained from the strip, which encircles two corners; hence $z(\alpha_0 * \alpha_1) \equiv 2(k - k')$. $\square$

Proposition 4.15. *Put $\deg = \text{gr} - \sigma(K_q)$, normalized by $\text{gr}(r_+) = \sigma(K_q)$ as in (2.5). For a crossing P of $\hat{A}$ with $\mathcal{D}_k$,*

$$\deg x_P^+ = \begin{cases} 2k, & P \text{ an up-crossing,} \\ 2k + 1, & P \text{ a down-crossing,} \end{cases} \quad \deg x_P^- = \deg x_P^+ - 1.$$

On the remnant circle the same formula holds for the relative grading: for crossings P, P' of $\hat{C}$ with $\mathcal{D}_k, \mathcal{D}_{k'}$, $\deg x_P^+ - \deg x_{P'}^+ \equiv 2(k - k') + \delta_P - \delta_{P'}$, with $\delta_P = 1$ for down-crossings and 0 for up-crossings.

Proof. On the arc, apply Lemma 4.14 to $p = x_P^+ \in S_+^k$ and $p' = r_+ \in S_+^0$ along $\hat{A}$. By Proposition 4.11(3), μ counts $+1$ for each U_{J_2} and -1 for each U_{J_3} passed. Up-crossings lie before U_{J_2} or after U_{J_3} , so $\mu = 0$; down-crossings lie between them, so $\mu = +1$. Then $\text{gr}(x_P^+, x_P^-) = 1$ [HHK18, §5.1]. On the circle the same argument applies along its lift: between two up-crossings the neighborhoods U_{J_2} and U_{J_3} alternate, so their net count is 0, and from an up-crossing to the next down-crossing it is $+1$. Over one period $\mu = 0$, $z \equiv 0$ and k changes by 2, which is consistent. $\square$

4.7. Bigons. A bigon lifts to an immersed disc in $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2$, since $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2 \rightarrow P^*$ is a covering, and by the first statement of (P9) its boundary is a sub-path of one lift of a component of L_1 together with an arc of one lift of L_0 . The following lemma counts such discs in the configuration of Figure 1(c). Each lift $S_{\pm}^k$ of L_0 is a graph over γ and leaves the open strip at both ends; a transverse crossing with it is an up- or down-crossing according as the curve passes from below $S_{\pm}^k$ to above it or conversely.

Lemma 4.16. *Let $\tilde{L}$ be an embedded lift of a component of L_1 that lies in the open strip $\{0 < \gamma < \pi\}$, apart from endpoints at lattice points, and is either an arc or a properly embedded line. Let Σ be one of the lifts $S_{\pm}^k$ of L_0 , and suppose that $\tilde{L} \cap \Sigma = \{p_0, p_1, \dots, p_{2d}\}$, all transverse and in this order along $\tilde{L}$, with p_{2i} up-crossings and p_{2i+1} down-crossings, and with $\gamma(p_0) > \gamma(p_1) > \dots > \gamma(p_{2d})$. Then the bigons whose boundary lifts to a sub-path of $\tilde{L}$ and an arc of Σ are exactly $2d$ bigons, one from p_{2i-1} to p_{2i-2} and one from p_{2i-1} to p_{2i} for each $1 \leq i \leq d$; each of them lifts to an embedded disc in $\mathbb{R}^2$.*

Proof. For $0 \leq s < 2d$ let Γ_s be the union of the sub-path $\tilde{L}[p_s, p_{s+1}]$ and the arc $\Sigma[p_{s+1}, p_s]$. The first meets Σ only at its ends, because the points are consecutive along $\tilde{L}$, and the second contains no point of $\tilde{L}$ in its interior, because they are consecutive in the order of γ . So Γ_s is a Jordan curve; let D_s be its bounded complementary component. The two closed half-planes complementary to the strip are connected, unbounded and disjoint from Γ_s , so D_s lies in the strip and contains no lattice point. Between consecutive crossings the curve stays on one side of Σ , so D_s lies above Σ if p_s is an up-crossing and below Σ if it is a down-crossing: the domains alternate. They are pairwise disjoint, since their boundaries meet only at shared endpoints and $\tilde{D}_s \cap \Sigma = \Sigma[p_{s+1}, p_s]$.

Existence. At the corners of D_s the four continuations leave D_s without entering it: the rays of $\tilde{L}$ beyond p_s and p_{s+1} meet Γ_s nowhere else, by embeddedness and the order of γ , and run

to lattice points outside the strip or to infinity; the rays of Σ leave the strip. So both corners are convex, and D_s projects to a bigon. If D_s lies above Σ , its counterclockwise boundary runs along Σ in the direction of increasing γ , from p_{s+1} to p_s ; if below, from p_s to p_{s+1} . In both cases the bigon runs from the down-crossing to the up-crossing, in agreement with Proposition 4.15 and [HHK18, Prop. 3.8].

Uniqueness. The L_1 -side of a bigon as in the statement is $\tilde{L}[p_i, p_j]$ for some $i \neq j$, and its L_0 -side is $\Sigma[p_j, p_i]$, since Σ is an embedded line. If $|i - j| \geq 2$, its boundary is, as a 1-cycle, the sum of the $|i - j|$ consecutive curves Γ_s between p_i and p_j , each traversed with its sub-path of $\tilde{L}$ in the same direction. Since the domains alternate above and below Σ , consecutive curves Γ_s are traversed with opposite orientations, so the boundary winds -1 times around the points of one of the domains. This is impossible for the boundary of an orientation-preserving immersed disc, whose winding number around a point is its number of preimages. If $|i - j| = 1$ the boundary is Γ_s , and only the direction from the down-crossing is counterclockwise. There is at most one bigon from a given generator to another [HHK18, Cor. 3.13]. $\square$

Proposition 4.17. (1) *In the joined case the complex is the direct sum, over the lifts $S_{\pm}^k$ of L_0 , of subcomplexes $C_k^{\pm}$. If $\mathcal{D}_k$ is crossed once, $C_k^{\pm}$ has one generator. If $\mathcal{D}_k$ is crossed three times, at P_1 (up), P_2 (down) and P_3 (up), then $\partial x_{P_2}^{\pm} = x_{P_1}^{\pm} + x_{P_3}^{\pm}$, and $H(C_k^{\pm}) = \mathbb{F}_2$, generated by $[x_{P_1}^{\pm}]$ in degree $2k$ for the sign $+$ and $2k - 1$ for the sign $-$. There are no other bigons; in particular none involves r_+ . The differential has rank $2N_q$.*

(2) *In the split case the arc carries no bigon. The complex of the remnant circle is the direct sum of four chains, one for each sign and each residue of k modulo 2; each has generators $p_0, \dots, p_{2d}$ with $\partial p_{2i-1} = p_{2i-2} + p_{2i}$, and homology $\mathbb{F}_2$ in the degree of its up-crossings. The circle contributes rank 1 in each of the four degrees, for every choice of absolute grading on it; its differential has rank $2N_q$.*

Proof. By the first statement of (P9) the boundary of a bigon lifts to a sub-path of one lift of a component of L_1 and an arc of one lift of L_0 . For the arc there is one lift through each generator, $\hat{A}$. The only elements of G preserving the open strip are the vertical translations, so for the circle a sub-path of a lift is unique up to the translation $T = (0, 4\pi)$, which carries $S_{\pm}^k$ to $S_{\pm}^{k+2}$.

(1) On a lift of L_0 met three times, the order along $\hat{A}$ is P_1 (near $S_{-m}(H)$), P_2 (near $S_{-m}(\Omega_1^{-1})$), P_3 (near $S_{-m}(H')$), because $\varphi_1 < 2\pi k < \varphi_2 < \varphi_4$. At $\varepsilon = 0$ the abscissae of the three crossings are

$$\gamma_1 = \frac{(2k+1)\pi}{q}, \quad \gamma_2 = \frac{2\pi(k-2)}{q-6}, \quad \gamma_3 = \frac{(2k-3)\pi}{q},$$

from $q\gamma - \pi = 2\pi k$ on $S_{-m}(H)$, $4\pi + (q-6)\gamma = 2\pi k$ on $S_{-m}(\Omega_1^{-1})$ and $q\gamma + 3\pi = 2\pi k$ on $S_{-m}(H')$. With $x = 2\pi(k-2)$,

$$\gamma_2 - \gamma_3 = \frac{6(x - \frac{(q-6)\pi}{6})}{q(q-6)}, \quad \gamma_1 - \gamma_2 = \frac{6(\frac{5(q-6)\pi}{6} - x)}{q(q-6)}.$$

Since $\varphi_3 < 2\pi k < \varphi_2$ and both $\varphi_2 - 2\pi k$ and $2\pi k - \varphi_3$ are at least $\pi/6$, both differences are at least $\pi/(q(q-6))$. So for small perturbations $\gamma(P_1) > \gamma(P_2) > \gamma(P_3)$, and Lemma 4.16 applies with $d = 1$. A lift of L_0 met once carries one generator, and $S_{\pm}^0$ carries only r_+ , since $\varphi > 0$ after the corner. The generators on $S_{\pm}^k$ and on $S_{\pm}^k$ have degrees $(u, u+1, u)$ with $u = 2k$, resp. $u = 2k-1$ (Proposition 4.15). There are N_q such lines, each with two lifts of L_0 near it, so the differential has rank $2N_q$.

(2) The arc is φ -monotone, so it carries no bigon, by the argument of the second statement of (P9) applied to $\hat{A}$, using Proposition 4.11(2) and, as in the proof of Proposition 4.13, the fact that no crossing with a line $\mathcal{D}_k$ lies in a junction neighborhood. Along the lift of the circle the pieces

alternate, dec_j near $S_{-m}(\Omega_1^{-1}) + (0, 4\pi j)$ and inc_j near $S_{-m}(H') + (0, 4\pi j)$, and each crosses a given line $\mathcal{D}_k$ at most once. By Proposition 4.11, the range of φ on dec_j is the intersection of its ranges on inc_{j-1} and inc_j . Hence the crossings of $\mathcal{D}_k$ along the lift alternate between up and down, beginning and ending with an up-crossing (as $\varphi \rightarrow \pm\infty$ along the lift), and a down-crossing on dec_j has the up-crossings on inc_{j-1} and inc_j as neighbors. The triple $(\text{inc}_{j-1}, \text{dec}_j, \text{inc}_j)$ is the triple $(S_{-m}(H), S_{-m}(\Omega_1^{-1}), S_{-m}(H'))$ of the arc translated by $(0, 4\pi j)$, so the abscissa computation above shows that γ decreases strictly along the sequence, and Lemma 4.16 applies to every lift of L_0 . The complex of the circle in P^* is the quotient by $\langle T \rangle$: its generators are the T -orbits of crossing points, represented exactly once on the four lifts $S_{\pm}^{k_0}$ with $k_0 \in \{0, 1\}$, and its bigons are the T -orbits of the domains of Lemma 4.16. Each chain has homology $\mathbb{F}_2$ in the degree of its up-crossings. By Proposition 4.15 these degrees are $2k_0$ and $2k_0 - 1$ up to a common shift, that is, 0, 2, 3, 1. Finally $4 + 4N_q = \sum(2d + 1)$ over the four chains, so $\sum d = 2N_q$ is the rank of the differential. $\square$

4.8. Proof of Theorem 1.2.

Theorem 4.18. *Let $q \geq 7$ be odd and $a > 0$. For the decomposition $K_q = (Y, T) \cup_{t_e^{-(q-5)/2}} (D, U)$, every sufficiently small $\varepsilon \in \mathcal{C}_a \setminus \{0\}$, the threshold depending on q , and every sufficiently small earring parameter:*

- (1) L_1 is a restricted immersed 1-manifold, (L_0, L_1) is admissible, and $C(L_0, L_1)$ has $q + 2 + 4N_q$ generators;
- (2) the differential has rank $2N_q$ and is described by Proposition 4.17; it is nonzero for $q \geq 9$, and no bigon involves r_+ ;
- (3) with $\text{gr}(r_+) = \sigma(K_q)$, $H^\natural(Y, T, \pi)$ has rank $q + 2$ and graded dimensions $(1 + N_3, N_1, N_1, N_3)$ in degrees $\text{gr} - \sigma(K_q) = 0, 1, 2, 3$, $N_1 = \lceil \frac{q+1}{4} \rceil$, $N_3 = \lfloor \frac{q+1}{4} \rfloor$.

This holds in both resolutions, and for every choice of absolute grading on the remnant circle.

Proof. (1) By Proposition 4.11(1) and Lemma 4.6(c), the arc is a proper immersion with embedded lift, so it has no fishtail and is a restricted immersed arc; the remnant circle is restricted by Proposition 4.11. Transversality is Proposition 4.13. Admissibility follows as in the proof of [Wue26c, Lemma 6.6] (see (P9)): loops on L_0 have classes in the span of $e_{(0,0)}$ and $e_{(\pi,\pi)}$ in $H_1(P^*)$, while a loop running j times around the remnant circle, which is homotopic in the strip to twice the core, has class $\pm 2j(e_{(0,0)} + e_{(0,\pi)})$, which lies in that span only if $j = 0$. The generator count is Proposition 4.13.

(2) This is Proposition 4.17, with $N_q \geq 1$ for $q \geq 9$.

(3) In the joined case, sum Proposition 4.17(1) over $k = 1, \dots, k_{\max}$ with $k_{\max} = \frac{q+1}{2}$, and add r_+ in degree 0. The lift S_+^k contributes degree $2k$ and S_-^k degree $2k - 1$; modulo 4, odd k gives degrees 2 and 1, and even k gives 0 and 3. So the homology is $(1 + \lfloor k_{\max}/2 \rfloor, \lceil k_{\max}/2 \rceil, \lceil k_{\max}/2 \rceil, \lfloor k_{\max}/2 \rfloor)$, and $\lceil k_{\max}/2 \rceil = N_1$, $\lfloor k_{\max}/2 \rfloor = N_3$. In the split case the arc crosses $\mathcal{D}_k$ once for $k = 1, \dots, k_{\max} - 2$, which gives $(\lfloor k_{\max}/2 \rfloor, \lceil k_{\max}/2 \rceil - 1, \lceil k_{\max}/2 \rceil - 1, \lfloor k_{\max}/2 \rfloor - 1)$, and the circle adds $(1, 1, 1, 1)$ by Proposition 4.17(2). The sum is the same vector. $\square$

Proof of Theorem 1.2. By Theorem 4.4, $K_q = K^{(-m)}$ with $m = (q - 5)/2$, so K_q is the union of the tangle of [HHK18, §10] for $T(3, 5)$ with $(r, s) = (2, -1)$ and the trivial tangle, glued by t_e^{-m} ; this is a decomposition as in [HHK18, (17)], and by Lemma 4.6 its restriction map is $S_{-m} \circ L_1$. Theorem 4.18 gives restrictedness, admissibility, the generators and the differential, and the graded dimensions of $H^\natural(Y, T, \pi)$, with $\text{gr}(r_+) = \sigma(K_q) = -(q + 1)$ (Proposition 3.5). By Proposition 3.6, $I^\natural(K_q; \mathbb{F}_2)$ has the same graded dimensions in Kronheimer and Mrowka's grading, so $H^\natural(Y, T, \pi) \cong I^\natural(K_q; \mathbb{F}_2)$ as $\mathbb{Z}/4$ -graded vector spaces. This holds for every sufficiently

small perturbation in the cone $\mathcal{C}_a$, in either of its kinds of sector (Proposition 4.10), with zero bounding cochain; in particular Conjecture 2.5 holds for K_q . $\square$

For $q = 3$ and $q = 5$ the knots are $T(3, 4)$ and $T(3, 5)$, and the corresponding statement over $\mathbb{Q}$ is [Wue26c, Thm. 1.1, Cor. 1.2]. The knots K_q with $q \geq 7$ are hyperbolic and not two-bridge (Proposition 4.5).

Remark 4.19. At $\varepsilon = 0$ the intersection points of $S_{-m}(\Pi(W))$ with Δ are the restrictions of the traceless characters of K_q . The two halves H and H' of C_0 have the same image, their lifts differing by $(0, 4\pi)$, and the restriction map is 2-to-1 over the points of Δ in that image. So the irreducible traceless characters correspond to the crossings of the lifted sheared polygon of Corollary 4.8, rather than of its image, with the lines $\mathcal{D}_k$, and the generator count of Proposition 4.13 is one more than twice their number. It can be written $q + 2 + 4N_q = q - 6 + 4(\lfloor \frac{5q+6}{12} \rfloor - \lfloor \frac{q+6}{12} \rfloor)$, which is the count of [FKP17, §5, p. 29] read consistently: the rank formula printed there gives 13 at $q = 7$ (and the printed summands give 5), whereas the explicit count on p. 28 is 9. For $q \geq 9$ the traceless character variety therefore has more points than are needed to generate $I^\natural(K_q)$, and the complex is not minimal. This agrees with the remark of Daemi and Scaduto that a direct computation from the character variety shows that $P(-2, 3, 7)$ is I -basic, while such a computation does not work for $P(-2, 3, q)$ with $q > 7$ [DS24b, p. 161].

Remark 4.20. Put $\chi(x_{\bar{P}}) = x_P^+$ and $\chi(x_P) = \chi(r_+) = 0$. By Proposition 4.17 every bigon pairs $x_{P_2}^\pm$ with $x_{P_1}^\pm$ and $x_{P_3}^\pm$, for each sign, so the differential commutes with χ . The pillowcase complex is thus an S-complex over $\mathbb{F}_2$, in the sense of (3.4), with $\delta_1 = \delta_2 = 0$ and $v = 0$, and its irreducible part has a differential of rank N_q . The homology of the span of the generators $x_{\bar{P}}$ has one class for each line $\mathcal{D}_k$, $1 \leq k \leq \frac{q+1}{2}$, in degree $2k - 1$ (in the split case, with the remnant circle graded so that its generators x^- have odd degree, as in [Wue26c, Def. 6.13]): it has rank N_1 in degree 1 and N_3 in degree 3, the ranks of the irreducible instanton homology $I(K_q)$ of [DS24b, Prop. 8.30]. A chain-level comparison with $\tilde{C}(K_q; \mathbb{F}_2)$ is open. For torus knots the corresponding conjecture is formulated in [Wue26b, Conj. 1.5] and proved for $T(3, n)$ in [Wue26c, Thm. 1.3]; there the irreducible part has zero differential and the pillowcase differential is a component of type δ_1 , whereas for K_q with $q \geq 9$ it lies in the irreducible part.

Remark 4.21. The proofs above are by hand. Numerical computations of the perturbed curves, of their Floer complexes in both kinds of sector, and of the resolution type agree with Proposition 4.10, Proposition 4.15 and Theorem 4.18, apart from canceling pairs of generators that are missing at the larger perturbations, as in Remark 4.12; they are described in Appendix A, item (v), and are not used in the proofs.

5. TWISTED TORUS KNOTS

For $n \equiv 1, 2 \pmod{6}$ the traceless character variety of the tangle of [HHK18, §10] for $T(3, n)$ is, before and after a small holonomy perturbation, a disjoint union of an arc and $2\lfloor n/6 \rfloor$ circles, and the image of the unperturbed arc in the pillowcase is a straight segment [Wue26c, Props. 3.6, 3.9 and 5.7]. There is no central circle, so the argument of Section 4 goes through without an analogue of Lemma 4.7. Regluing the trivial tangle by powers of the twist t_e produces the twisted torus knots $T(3, n; 2, m)$, and Theorem 5.1 below computes their pillowcase homology, identifies the knots under a hypothesis on the chirality of the torus-knot decomposition, and compares the result with their instanton homology. On the pillowcase side, the one input beyond [Wue26c] and Section 4 is Hedden, Herald and Kirk's computation of the Floer homology of a circle that misses the edges [HHK18, Thm. 5.4], which we use for the closed components.

5.1. Statement. For $n \geq 1$ coprime to 3 and $m \geq 0$, the *twisted torus knot* $T(3, n; 2, m)$ is the closure of the positive braid

$$\beta_{n,m} = (\sigma_1 \sigma_2)^n \sigma_1^{2m} \in B_3,$$

that is, $T(3, n)$ with m positive full twists added on two adjacent strands; $T(3, n; 2, 0) = T(3, n)$. Clay and Watson write $T_{3,n}^m$ for this knot when $n \equiv 2 \pmod{3}$ [CW13, §3.2], and Daemi and Scaduto write $T(3, n; 2, m)$ [DS24b, §8.5]. The knots of Section 4 are the case $n = 5$: $T(3, 5; 2, m) = P(-2, 3, 5 + 2m)$. Throughout we put

$$\nu = \lfloor n/3 \rfloor, \quad M = \begin{cases} m, & n \equiv 1 \pmod{3}, \\ m + 1, & n \equiv 2 \pmod{3}, \end{cases}$$

and for $n \equiv 1, 2 \pmod{6}$ we write $n = 6j + 1$ or $n = 6j + 2$, so that $\nu = 2j$.

For $n \geq 4$ coprime to 3 let (Y_n, T_n) be the tangle of [HHK18, §10] for $T(3, n)$ with the parameters $(r, s) = ((2n + 1)/3, -2)$ if $n \equiv 1$ and $(r, s) = ((n + 1)/3, -1)$ if $n \equiv 2 \pmod{3}$, as in [Wue26c], and let $K^{(h)}(3, n) = (Y_n, T_n) \cup_{t_e^h} (D, U)$ be the regluing of Section 4; for $n = 5$ this is the knot $K^{(h)}$ of Section 4. Lemma 4.1 and its proof apply to every n , with the boundary words a, b, c of [Wue26c, (3.2)]: $\pi_1(S^3 \setminus K^{(h)}(3, n)) = \langle A, B \mid t_e^h(b) = c \rangle$ with $t_e^h(b) = (ba)^h b (ba)^{-h}$, and $K^{(h-1)}(3, n)$ differs from $K^{(h)}(3, n)$ by one full twist of two strands, hence by one crossing change. The knot group does not determine whether $K^{(0)}(3, n)$ is the positive torus knot $T(3, n)$ or its mirror image. For $n = 5$ we verified that it is $T(3, 5)$ (Remark 4.3); for the knots of Theorem 5.1 it is a hypothesis of the theorem, shared with [HHK18] and [Wue26c].

We write $\varepsilon = (\varepsilon_A, \varepsilon_B)$ for the parameters of the holonomy perturbation π of [Wue26c] (see (P1) in §4.3) and L_1^{tor} for the resulting curve of (Y_n, T_n) . By Lemma 4.6, the curve of (Y_n, T_n) in the decomposition of $K^{(-m)}(3, n)$ is $L_1 = S_{-m} \circ L_1^{\text{tor}}$, where $S_{-m}(\gamma, \theta) = (\gamma, \theta + 2m\gamma)$. For the trivial side we use the earring curve $L_0 = L_0^{\varepsilon, g}$ of [HHK18, Def. 3.4] with $\varepsilon > 0$ and $g = \delta \sin$; in [Wue26c], $\delta = 0$. Recall that $\ell(K)$ denotes the sum of the absolute values of the coefficients of Δ_K .

Theorem 5.1. *Let $n \geq 7$ with $n \equiv 1$ or $2 \pmod{6}$, let $m \geq 1$, and define j, ν, M as above. Assume, as in [HHK18] and [Wue26c], that the tangle (Y_n, T_n) glued to the trivial tangle, $K^{(0)}(3, n)$, is the positive torus knot $T(3, n)$. Put $K = K^{(-m)}(3, n)$ and $(M_0, M_1) = (\lfloor M/2 \rfloor, \lceil M/2 \rceil)$.*

- (i) $K = T(3, n; 2, m)$.
- (ii) $\sigma(K) = \sigma(T(3, n)) - 2m = -(4\nu + 2M)$, and $\ell(K) = |\sigma(K)| + 1$.
- (iii) *For every sufficiently small $\varepsilon \in \mathbb{R}^2 \setminus \{0\}$, L_1 is a restricted immersed 1-manifold consisting of an arc and $2j$ circles, and the circles miss the left and right edges and have vertical degree 2. There is $\epsilon_0 > 0$, depending on ε , such that for $0 < \epsilon < \epsilon_0$ and $|\delta| < \epsilon^2/2$ the arc meets L_0 transversely in $1 + 2M$ points and carries no bigon. For each such ϵ and almost every such δ the pair (L_0, L_1) is admissible, and with $\text{gr}(r_+) = \sigma(K)$ the graded dimensions of $H^\natural(Y_n, T_n, \pi)$ are*

$$(1 + 2j + M_0, 2j + M_1, 2j + M_1, 2j + M_0) \quad \text{in degrees } \text{gr} - \sigma(K) = 0, 1, 2, 3,$$

whatever absolute gradings are chosen on the circles.

- (iv) $I^\natural(K; \mathbb{Z})$ is free of rank $|\sigma(K)| + 1$, and $H^\natural(Y_n, T_n, \pi) \cong I^\natural(K; \mathbb{F}_2)$ as $\mathbb{Z}/4$ -graded vector spaces.

In particular, this decomposition of $T(3, n; 2, m)$ satisfies the conclusion of [HHK18, Conj. 6.5].

Remark 5.2. By Lemma 5.3 below, $T(3, 6j + 2; 2, m) = T(3, 6j + 1; 2, m + 1)$. The two families in Theorem 5.1 therefore consist of the knots $T(3, 6j + 1; 2, M)$ with $M \geq 1$, and the vector in (iii) depends only on (j, M) ; for $M \geq 2$ the knot arises from both tangles (Y_{6j+1}, T_{6j+1}) and (Y_{6j+2}, T_{6j+2}) . For $M = 1$ the knot $T(3, 6j + 1; 2, 1) = T(3, 6j + 2)$ is a torus knot, and

Theorem 5.1 gives a second decomposition of it, with the homology found in [Wue26c]. For $M \geq 2$ the knot is neither a torus knot, nor alternating, nor a pretzel knot $P(-2, 3, q)$ (Lemma 5.6).

5.2. Alexander polynomials. Let ψ be the reduced Burau representation of B_3 ,

$$\psi(\sigma_1) = \begin{pmatrix} -t & 1 \\ 0 & 1 \end{pmatrix}, \quad \psi(\sigma_2) = \begin{pmatrix} 1 & 0 \\ t & -t \end{pmatrix}.$$

For $\beta \in B_3$ whose closure $\hat{\beta}$ is a knot, $\Delta_{\hat{\beta}}(t) \doteq \frac{1-t}{1-t^3} \det(I - \psi(\beta))$ by Burau's formula [Bir74, Thm. 3.11]; here $\doteq$ denotes equality up to a unit $\pm t^k$. Daemi and Scaduto record a formula for the Alexander polynomials of the twisted torus links $T(p, q; 2, k)$ [DS24b, Prop. 8.32]; we need the coefficient norm, which the closed form below displays directly.

Lemma 5.3. *For $\nu, m \geq 0$, $\sigma_1^{-1}(\sigma_1\sigma_2)^{3\nu+2}\sigma_1^{2m}\sigma_1 = (\sigma_1\sigma_2)^{3\nu+1}\sigma_1^{2m+2}$ in B_3 . Hence $T(3, 3\nu + 2; 2, m) = T(3, 3\nu + 1; 2, m + 1)$ as oriented knots, and for every n coprime to 3 the knot $T(3, n; 2, m)$ is the closure of $(\sigma_1\sigma_2)^{3\nu+1}\sigma_1^{2M}$.*

Proof. The full twist $\Delta^2 = (\sigma_1\sigma_2)^3$ is central, and $(\sigma_1\sigma_2)^2 = \sigma_1(\sigma_2\sigma_1\sigma_2) = \sigma_1^2\sigma_2\sigma_1$ by the braid relation. Hence $(\sigma_1\sigma_2)^{3\nu+2}\sigma_1^{2m} = \Delta^{2\nu}\sigma_1^2\sigma_2\sigma_1^{2m+1}$, and conjugation by σ_1 gives $\Delta^{2\nu}\sigma_1\sigma_2\sigma_1^{2m+2} = (\sigma_1\sigma_2)^{3\nu+1}\sigma_1^{2m+2}$. Conjugate braids have isotopic closures. $\square$

For example, $P(-2, 3, 5 + 2m) = T(3, 5; 2, m) = T(3, 4; 2, m + 1)$ and $T(3, 6j + 1; 2, 1) = T(3, 6j + 2)$.

Proposition 5.4. *Let $\nu, M \geq 0$ and let $K_{\nu, M}$ be the closure of $(\sigma_1\sigma_2)^{3\nu+1}\sigma_1^{2M}$. Then*

$$\Delta_{K_{\nu, M}}(t) \doteq \underbrace{\sum_{i=0}^{\nu-1} (t^{3i} - t^{3i+1})}_{L(t)} + \underbrace{\sum_{i=0}^{2M} (-1)^i t^{3\nu+i}}_{L_{\text{mid}}(t)} + \underbrace{\sum_{i=0}^{\nu-1} (t^{6\nu+2M-3i} - t^{6\nu+2M-3i-1})}_{L^*(t)}.$$

The three sums have disjoint exponent ranges, every coefficient is 0 or ± 1 , the nonzero coefficients alternate in sign, and $\ell(K_{\nu, M}) = 4\nu + 2M + 1$.

Proof. Put $P = \psi(\sigma_1\sigma_2) = \begin{pmatrix} 0 & -t \\ t & -t \end{pmatrix}$ and $T = t^{2M}$. Since $\text{tr } P = -t$ and $\det P = t^2$, the Cayley–

Hamilton theorem gives $P^3 = t^3 I$, and induction on M gives $\psi(\sigma_1)^{2M} = \begin{pmatrix} T & (1-T)/(1+t) \\ 0 & 1 \end{pmatrix}$.

The matrix $X = t^{3\nu} P \psi(\sigma_1)^{2M}$ has $\text{tr } X = -t^{3\nu+1}(t + T)/(1 + t)$ and $\det X = t^{6\nu+2}T$, so $\det(I - X) = 1 - \text{tr } X + \det X$ is

$$F_{\nu, M}(t) := \det(I - \psi((\sigma_1\sigma_2)^{3\nu+1}\sigma_1^{2M})) = 1 + t^{3\nu+2} \frac{1 + t^{2M-1}}{1 + t} + t^{6\nu+2M+2} \quad (5.1)$$

(for $M = 0$ the middle term is $t^{3\nu+1}$). Since $(1-t)/(1-t^3) = 1/(1+t+t^2)$, it suffices to show that $(1+t+t^2)(L + L_{\text{mid}} + L^*) = F_{\nu, M}$. From $(1+t+t^2)(1-t) = 1-t^3$ and $L^*(t) = t^{6\nu+2M}L(t^{-1})$,

$$(1+t+t^2)L = 1 - t^{3\nu}, \quad (1+t+t^2)L^* = t^{6\nu+2M+2} - t^{3\nu+2M+2}.$$

Next, $L_{\text{mid}} = t^{3\nu}(1+t^{2M+1})/(1+t)$ and $(1+t+t^2)(1+t^{2M+1}) - (1+t)(1+t^{2M+2}) = t^2 + t^{2M+1}$, so

$$(1+t+t^2)L_{\text{mid}} = t^{3\nu} + t^{3\nu+2M+2} + t^{3\nu+2} \frac{1 + t^{2M-1}}{1 + t}.$$

The three products add up to $F_{\nu, M}$. The sums L , L_{mid} and L^* occupy the exponents $0, \dots, 3\nu - 2$, then $3\nu, \dots, 3\nu + 2M$, then $3\nu + 2M + 2, \dots, 6\nu + 2M$, so no cancellation occurs, and they contribute 2ν , $2M + 1$ and 2ν nonzero coefficients. Inside each sum the signs alternate; the last coefficient of L and the first of L^* are -1 , and L_{mid} begins and ends with $+1$. $\square$

The degree $6\nu + 2M = 2(n + m - 1)$ is twice the genus of the positive braid closure. For $T(3, 7; 2, 2)$, where $\nu = M = 2$, the coefficients of $t^0, \dots, t^{16}$ are

$$1, -1, 0, 1, -1, 0, 1, -1, 1, -1, 1, 0, -1, 1, 0, -1, 1,$$

and $\ell = 13 = |\sigma| + 1$ (Proposition 5.5).

5.3. Signature.

Proposition 5.5. *Let $n \equiv 1, 2 \pmod{6}$ and $m \geq 0$. Then $\det T(3, n; 2, m) = 2m + 1$ for $n \equiv 1 \pmod{6}$ and $2m + 3$ for $n \equiv 2 \pmod{6}$, and*

$$\sigma(T(3, n; 2, m)) = \sigma(T(3, n)) - 2m = -(4\nu + 2M).$$

Consequently $\ell(T(3, n; 2, m)) = |\sigma(T(3, n; 2, m))| + 1$.

Proof. At $t = -1$, $\psi(\sigma_1) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\psi(\sigma_2) = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$, so $Q = \psi(\sigma_1 \sigma_2)|_{t=-1} = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}$, which has order 6. Since $1 + t + t^2 = 1$ at $t = -1$, $\det T(3, n; 2, m) = |\det(I - Q^n \psi(\sigma_1)^{2m})|_{t=-1}|$. For $n \equiv 1 \pmod{6}$ the matrix $Q^n \psi(\sigma_1)^{2m}|_{t=-1}$ is $\begin{pmatrix} 0 & 1 \\ -1 & 1 - 2m \end{pmatrix}$, and $\det \begin{pmatrix} 1 & -1 \\ 1 & 2m \end{pmatrix} = 2m + 1$; for $n \equiv 2 \pmod{6}$ it is $\begin{pmatrix} -1 & 1 - 2m \\ -1 & -2m \end{pmatrix}$, and $\det \begin{pmatrix} 2 & 2m - 1 \\ 1 & 2m + 1 \end{pmatrix} = 2m + 3$.

By Murasugi's theorem [Mur65, Thm. 5.6], if $|\sigma(K)| \equiv 2\mu \pmod{4}$ then $|\Delta_K(-1)| \equiv (-1)^\mu \pmod{4}$. As m increases by one, the determinant alternates between 1 and 3 modulo 4, so the parity of $\sigma/2$ changes and $\sigma(T(3, n; 2, m + 1)) \neq \sigma(T(3, n; 2, m))$. Changing one crossing of σ_1^{2m+2} in $\beta_{n, m+1}$ from positive to negative gives $\beta_{n, m}$. For knots $K_\pm$ related by such a change, $\sigma(K_+) - \sigma(K_-) \in \{0, -2\}$ in our convention (the standard Seifert-matrix computation; see [DS24b, p. 163]). Hence $\sigma(T(3, n; 2, m + 1)) = \sigma(T(3, n; 2, m)) - 2$. Finally $|\sigma(T(3, n))| = 2(n - 1) - 4\lfloor n/6 \rfloor$ [Lit79] (see [Wue26c, proof of Prop. 6.3]), which is 4ν for $n = 6j + 1$ and $4\nu + 2$ for $n = 6j + 2$. The last assertion follows from Proposition 5.4 and Lemma 5.3. $\square$

For $n \equiv 2 \pmod{6}$ this agrees with the formula $\sigma(T(3, 3n' + 2; 2, k)) = -4n' - 2 - 2k$, $k > 0$, of [DS24b, p. 163]. The argument needs the determinant to alternate modulo 4; for $n = 5$ it does not at the first step, and indeed $\sigma(P(-2, 3, 5)) = \sigma(P(-2, 3, 7)) = -8$.

Lemma 5.6. *Let $\nu \geq 2$ and $M \geq 2$. Then $K_{\nu, M}$ is not a torus knot, not alternating, and not a pretzel knot $P(-2, 3, q)$.*

Proof. By Proposition 5.4, $\Delta_{K_{\nu, M}} = 1 - t + t^3 - \dots$, with zero coefficient of t^2 . As in the proof of Lemma 3.3, a torus knot with this Alexander polynomial would be $T(3, b)$ with $2(b - 1) = 6\nu + 2M$. By Lemma 5.3, $T(3, b)$ is $K_{\nu', 0}$ or $K_{\nu', 1}$ for some ν' , so by Proposition 5.4 its Alexander polynomial has no more than three consecutive nonzero coefficients, whereas $\Delta_{K_{\nu, M}}$ has $2M + 1 \geq 5$. Next, $\det K_{\nu, M} = |F_{\nu, M}(-1)| = |2 + (-1)^\nu(2M - 1)| < 4\nu + 2M + 1 = \ell(K_{\nu, M})$, while for an alternating knot the determinant equals ℓ (see the proof of Lemma 3.3). Finally, the coefficient of t^5 in $\Delta_{K_{\nu, M}}$ is 0, because $5 = 3 \cdot 1 + 2$ is not an exponent of L and L_{mid} begins at $3\nu \geq 6$, whereas in $\Delta_{P(-2, 3, q)}$ it is 1 for $q \geq 5$ (Lemma 3.1); and $P(-2, 3, q)$ is a torus knot for $q \leq 5$. $\square$

5.4. Identification of the knots. Clay and Watson prove that for $\nu \geq 0$ and $m \geq 0$ the group of $T(3, 3\nu + 2; 2, m) = T_{3, 3\nu+2}^m$ is

$$\pi_{3\nu+2}^m = \langle \alpha, \beta \mid \alpha^2(\beta^{-\nu}\alpha)^m\alpha = \beta^{2\nu+1}(\beta^{-\nu}\alpha)^m\beta^{\nu+1} \rangle$$

[CW13, §3.2 and Prop. 22].

- Lemma 5.7.** (a) If $n = 3\nu + 2 \geq 5$ and $m \geq 0$, then $\alpha \mapsto A$, $\beta \mapsto B^{-1}$ induces an isomorphism $\pi_{3\nu+2}^m \cong \pi_1(S^3 \setminus K^{(-m)}(3, n))$.
 (b) If $n = 3\nu + 1 \geq 4$ and $m \geq 1$, then $\alpha \mapsto AB^{2\nu+1}$, $\beta \mapsto B$ induces an isomorphism $\pi_{3\nu+2}^{m-1} \cong \pi_1(S^3 \setminus K^{(-m)}(3, n))$.

Proof. Both maps are automorphisms of the free group on two letters. The group of a one-relator presentation is unchanged, up to isomorphism, if the relator is replaced by its image under an automorphism of the free group, by a cyclic permutation, or by its inverse. So it suffices to show that each map carries Clay and Watson's relator $\alpha^2 Y^m \alpha \beta^{-(\nu+1)} Y^{-m} \beta^{-(2\nu+1)}$, where $Y = \beta^{-\nu} \alpha$ (with $m-1$ in place of m in (b)), to a cyclic permutation of the relator, or of its inverse, of the presentation $\langle A, B \mid (ba)^{-m} b (ba)^m = c \rangle$.

(a) Here $a = A^2 B^{2\nu+1}$, $b = B^{-(\nu+1)} A^{-1}$ and $c = B^{-(\nu+1)} a B^{\nu+1}$ [Wue26c, (3.2)]. With $C = AB^\nu$ we have $ba = B^{-(\nu+1)} C B^{\nu+1}$, so the relation reads $C^{-m} A^{-1} B^{-(\nu+1)} C^m = A^2 B^{2\nu+1}$, a cyclic permutation of the inverse of the corresponding relator is $R = AC^m A^2 B^{2\nu+1} C^{-m} B^{\nu+1}$. Under $\alpha \mapsto A$, $\beta \mapsto B^{-1}$ we have $Y \mapsto B^\nu A$, $A(B^\nu A)^m = C^m A$ and $(B^\nu A)^{-m} = B^\nu C^{-m} B^{-\nu}$, so Clay and Watson's relator goes to $AC^m A^2 \cdot B^{\nu+1} \cdot B^\nu C^{-m} B^{-\nu} \cdot B^{2\nu+1} = R$.

(b) Here $a = AB^\nu$, $b = B^{-(2\nu+1)} A^{-2}$ and $c = B^{-(2\nu+1)} AB^{3\nu+1}$ [Wue26c, (3.2)]. With $\Xi = B^{\nu+1} A$ we have $ba = B^{-(2\nu+1)} \Xi^{-1} B^{2\nu+1}$, so the relation reads $\Xi^m A^{-2} B^{-(2\nu+1)} \Xi^{-m} = AB^\nu$, whose relator has inverse $R_H = AB^\nu \Xi^m B^{2\nu+1} A^2 \Xi^{-m}$. Under $\alpha \mapsto AB^{2\nu+1}$, $\beta \mapsto B$ we have $Y \mapsto B^{-\nu} AB^{2\nu+1} = B^{-(2\nu+1)} \Xi B^{2\nu+1}$. Using $B^{2\nu+1} A = B^\nu \Xi$, the left side $\alpha^2 Y^{m-1} \alpha$ goes to $AB^\nu \Xi^m B^\nu \Xi B^{2\nu+1}$ and the right side $\beta^{2\nu+1} Y^{m-1} \beta^{\nu+1}$ to $\Xi^{m-1} B^{3\nu+2}$. The relator therefore goes to $AB^\nu \Xi^m B^\nu \Xi B^{-(\nu+1)} \Xi^{1-m}$, which equals R_H because $B^\nu \Xi = B^{2\nu+1} A$ and $AB^{-(\nu+1)} \Xi = A^2$. $\square$

At $m = 0$ in (a), and at $m = 1$ in (b), Clay and Watson's relation is $\alpha^3 = \beta^{3\nu+2}$, the group of the torus knot $T(3, 3\nu + 2)$; part (a) at $\nu = 1$ is Lemma 4.2.

Proof of Theorem 5.1 (i) and (ii). Let $K = K^{(-m)}(3, n)$. For $n = 3\nu + 2$, Lemma 5.7(a) identifies $\pi_1(S^3 \setminus K)$ with the group of $T(3, n; 2, m)$. For $n = 3\nu + 1$, part (b) identifies it with the group of $T(3, 3\nu + 2; 2, m - 1)$, which is $T(3, n; 2, m)$ by Lemma 5.3. Both groups are generated by two elements, and both knots are nontrivial by Proposition 5.4, so both knots are prime by Norwood's theorem [Nor82]. Prime knots with isomorphic groups are equivalent [GoLu89, Cor. 2.1], by a homeomorphism of S^3 that may reverse orientation. Hence K is $T(3, n; 2, m)$ or its mirror image.

The chirality follows by induction on m , as in the proof of Theorem 4.4. By hypothesis $K^{(0)}(3, n) = T(3, n)$. Suppose that $\sigma(K^{(-m)}(3, n)) = \sigma(T(3, n)) - 2m$ for some $m \geq 0$. The knot $K^{(-m-1)}(3, n)$ differs from $K^{(-m)}(3, n)$ by one crossing change, so their signatures differ by at most 2 [Mur65, (10.2)]. By Proposition 5.5, $T(3, n; 2, m + 1)$ and its mirror image have signatures $\mp s$ with $s = |\sigma(T(3, n))| + 2m + 2$, and only $-s$ lies within 2 of $\sigma(T(3, n)) - 2m = -(s - 2)$, since $s + (s - 2) > 2$. Hence $K^{(-m-1)}(3, n) = T(3, n; 2, m + 1)$. This proves (i), and (ii) is Proposition 5.5. $\square$

5.5. The pillowcase homology.

Proof of Theorem 5.1 (iii). Let $\omega = 2$ for $n \equiv 1$ and $\omega = 4$ for $n \equiv 2 \pmod{6}$, so that $\omega + 2m - 1 = 2M + 1$.

The arc. By [Wue26c, Prop. 5.7(3) and Lemma 6.1], for small $\varepsilon \neq 0$ the lift $\tilde{A}^{\text{tor}}$ of the arc of L_1^{tor} starting at $(0, 0)$ is, away from its ends, C^1 -close to the segment from $(0, 0)$ to $(\pi, \omega\pi)$, along which $\theta = \omega\gamma$ [Wue26c, Prop. 3.9]. Near $(0, 0)$ it has the form $y \mathbf{c}(\varepsilon) + y^3 R(y, \varepsilon)$, where $\mathbf{c} = (c_1, c_2)$ and R are real-analytic and $\mathbf{c}(0)$ is a positive multiple of $(1, \omega)$ [Wue26c, Lemma 5.1], and the same holds at its other end. The shear S_{-m} is linear. Hence the lift $\tilde{A} = S_{-m} \tilde{A}^{\text{tor}}$ is C^1 -close, away from its ends, to the segment from $(0, 0)$ to $(\pi, (\omega + 2m)\pi)$, on which $\varphi = (2M + 1)\gamma$; near

$(0, 0)$ it equals $y S_{-m} \mathbf{c}(\varepsilon) + y^3 S_{-m} R(y, \varepsilon)$, and the φ -component of $S_{-m} \mathbf{c}(0)$ is $(2M+1) c_1(0) > 0$. So, as in [Wue26c, Cor. 5.3], φ is strictly monotone along $\tilde{A}$ with $|d\varphi|$ bounded below in arclength, and the limiting slopes $\omega + 2m + O(|\varepsilon|)$ differ from 1. The interior of $\tilde{A}$ lies in the open strip $0 < \gamma < \pi$, and $\tilde{A}$ ends on the line $\varphi = (2M+1)\pi$. It therefore crosses exactly the lines $\mathcal{D}_1, \dots, \mathcal{D}_M$, each once and transversally, and meets $\mathcal{D}_0$ only at its initial point. Since $\tilde{A}$ is embedded, the arc has no fishtail and is a restricted immersed arc.

The circles. By [Wue26c, Prop. 5.7(3)], the other components of L_1^{tor} are circles C^1 -close to the $2\lfloor \nu/2 \rfloor = 2j$ non-central circles [Wue26c, Prop. 3.6]. They are restricted, miss the left and right edges, and have vertical degree 2 [Wue26c, Cor. 4.8 and Lemma 6.6]. By Lemma 4.6 their images under S_{-m} have the same three properties. Along these images φ need not be monotone: S_{-m} adds $2m\gamma$ to φ , and γ is not monotone along a closed curve.

The earring curve. The lifts $S_{\pm}^k$ of $L_0 = L_0^{\varepsilon, g}$ are the translates of $\pm \tilde{L}_0^{\varepsilon, g}$ by $(0, 2\pi k)$, where $\tilde{L}_0^{\varepsilon, g}$ is given by [HHK18, (13)]. At the point with parameter t the slope of either is $(1 - \varepsilon \cos t)/(1 + \varepsilon \cos t) + \delta \cos \gamma$, and, for small ε , one has $\cos t > \varepsilon/2$ on S_+^k over the front strip $0 < \gamma < \pi$, and $\cos t < -\varepsilon/2$ on S_-^k over $2\pi \leq \gamma \leq \pi - 2\varepsilon$. Hence for $|\delta| < \varepsilon^2/2$ the lifts S_+^k have slope less than 1 over the front strip and the lifts S_-^k slope greater than 1 over $2\pi \leq \gamma \leq \pi - 2\varepsilon$, as for $\delta = 0$ [Wue26c, §2.2], and they lie within $\sqrt{2}\varepsilon + |\delta|$ of $\mathcal{D}_k$. The proofs of [Wue26c, Cor. 5.3(4), Prop. 6.3 and Lemma 6.8] use only these properties of the lifts of L_0 , the slope of S_-^k only at points at distance much larger than ε from the edges, and the properties of the arc established above. Therefore, for small ε and $|\delta| < \varepsilon^2/2$, the arc meets L_0 transversely in exactly $1 + 2M$ points: the generator r_+ of [HHK18, Lemma 5.1], which lies on S_+^0 because $\tilde{A}$ leaves $(0, 0)$ above $\mathcal{D}_0$, and a pair $x_k^{\pm} \in S_{\pm}^k$ near the crossing of $\tilde{A}$ with $\mathcal{D}_k$ for $k = 1, \dots, M$.

Transversality on the circles. Fix ε and let C be a circle of L_1 . Since $L_0^{\varepsilon, \delta \sin} = c_{\delta \sin} \circ L_0^{\varepsilon, 0}$, where $c_g(\gamma, \theta) = (\gamma, \theta + g(\gamma))$ [HHK18, (11), p. 18], $L_0^{\varepsilon, \delta \sin}$ is transverse to C if and only if $L_0^{\varepsilon, 0}$ is transverse to $c_{-\delta \sin}(C)$. Consider $\Phi(t, x, \delta) = \tilde{L}_0(t) - c_{-\delta \sin}(\tilde{C}(x)) \in \mathbb{R}^2$, where $\tilde{L}_0$ is a lift of $L_0^{\varepsilon, 0}$ and $\tilde{C}$ a lift of C . Its derivative in t is the tangent vector of $\tilde{L}_0$, which is never vertical, and its derivative in δ is $(0, \sin \gamma(\tilde{C}(x)))$, which is vertical and nonzero because C misses the edges. So Φ is a submersion, $\Phi^{-1}(0)$ is a 1-manifold, and δ is a regular value of the projection $\Phi^{-1}(0) \rightarrow \mathbb{R}$ exactly when 0 is a regular value of $\Phi(\cdot, \cdot, \delta)$. By Sard's theorem, applied to the countably many pairs of lifts, $L_0^{\varepsilon, \delta \sin}$ is transverse to every circle of L_1 for almost every δ . For such δ the pair (L_0, L_1) is admissible: L_0 is a circle, the intersections are transverse, and the argument of [Wue26c, Lemma 6.6] applies, since L_0 lies near Δ and each circle of L_1 lies in the open strip with vertical degree 2.

Bigons and gradings on the arc. By [Wue26c, Lemma 6.4] ((P9) in §4.4), the boundary of a bigon from p to p' lifts to a closed curve made of a sub-path of one lift of a component of L_1 and an arc of one lift of L_0 , which contains lifts $\tilde{p}, \tilde{p}'$ of both corners. Every lift of the arc is a translate of $\tilde{A}$ under the deck group G , which permutes the lifts of L_0 , and $\tilde{A}$ meets each of them at most once. So a bigon between generators on the arc would have $\tilde{p} = \tilde{p}'$, hence $p = p'$, which is impossible since $\text{gr}(p, p') = 1$ [HHK18, Prop. 3.8]. Thus the arc carries no bigon. For the degrees, $\deg = \text{gr} - \sigma(K)$ satisfies $\deg r_+ = 0$ by [HHK18, (19)] and $\deg x_k^- = \deg x_k^+ - 1$ [HHK18, §5.1]. Since φ has no critical point on $\tilde{A}$, [Wue26c, Lemma 6.8] applies to $r_+ \in S_+^0$ and $x_1^+ \in S_+^1$, and to x_{k-1}^+ and x_k^+ ; each step changes the degree by 2. Hence $\deg x_k^+ = 2k$ and $\deg x_k^- = 2k - 1$ modulo 4, and the homology of the arc, which equals its complex, has dimensions $(1 + M_0, M_1, M_1, M_0)$, since M_0 and M_1 count the even and the odd $k \in \{1, \dots, M\}$.

The total. By [HHK18, Thm. 5.4], each circle contributes $(1, 1, 1, 1)$, whatever its absolute grading; this applies to $L_0^{\varepsilon, \delta \sin} = c_{\delta \sin} \circ L_0^{\varepsilon, 0}$ because the pair $(L_0^{\varepsilon, \delta \sin}, C)$ is carried by $c_{-\delta \sin}$

to $(L_0^{\epsilon, 0}, c_{-\delta \sin}(C))$, and $c_{-\delta \sin}(C)$ again misses the edges and has vertical degree 2. There is no differential between different components [HHK18, p. 37], and $H^\natural(Y_n, T_n, \pi)$ is the direct sum of the contributions of the components [HHK18, p. 34]. Adding $2j$ copies of $(1, 1, 1, 1)$ to $(1 + M_0, M_1, M_1, M_0)$ gives the stated vector. $\square$

Remark 5.8. We have computed the sheared curves and their complexes numerically, by the method of [Vue26c, Rem. 6.12] with $\delta = 0$, for $n = 7, 8, 13$, $m = 1, 2, 3$, and the two perturbations $(n\varepsilon_A, \varepsilon_B, \epsilon) = (0.3, 0.05, 0.01)$ and $(-0.3, -0.03, 0.02)$. In all 18 cases the arc carries $1 + 2M$ generators and no bigon, the homology of each circle, computed from its complex rather than from [HHK18, Thm. 5.4], is $(1, 1, 1, 1)$, and the total is the vector of Theorem 5.1(iii). These computations are not used in the proofs.

5.6. Comparison with instanton homology.

Proposition 5.9. *Let $n \geq 7$ with $n \equiv 1, 2 \pmod{6}$, let $m \geq 0$, and let $K = T(3, n; 2, m)$. Then $I^\natural(K; \mathbb{Z})$ is free of rank $|\sigma(K)| + 1$, and in Kronheimer and Mrowka's absolute grading its graded ranks are*

$$(1 + \lfloor |\sigma(K)|/4 \rfloor, \lceil |\sigma(K)|/4 \rceil, \lceil |\sigma(K)|/4 \rceil, \lfloor |\sigma(K)|/4 \rfloor) \quad \text{in degrees } \text{gr} - \sigma(K) = 0, 1, 2, 3.$$

The same holds with coefficients in $\mathbb{Q}$ and in $\mathbb{F}_2$.

Proof. Since $n = 3(2j + 1) - 2$ or $n = 3 \cdot 2j + 2$, and $\det K \neq 0$ by Proposition 5.5, K is I -basic by [DS24b, Prop. 8.31], applied with $p = 3$ and with m full twists: its irreducible instanton homology $I(K)$ has rank $|\sigma(K)|/2$. The proof of [DS24b, Prop. 8.31] applies [DS24b, Lemma 8.23] starting from $T(3, n)$, whose irreducible complex has rank $|\sigma(T(3, n))|/2$ equal to the absolute value of its Euler characteristic [DS24b, Ex. 8.11], so that $I(T(3, n))$ is free abelian; by [DS24b, Rem. 8.27], $I(K)$ is free abelian too. By [DS24b, Prop. 8.28], $I(K) \cong \mathbb{Z}_{(1)}^{\lceil |\sigma|/4 \rceil} \oplus \mathbb{Z}_{(3)}^{\lfloor |\sigma|/4 \rfloor}$ in the grading of (3.4). For $n \equiv 2 \pmod{6}$ and $m \geq 1$ this is [DS24b, Cor. 8.36], with their n equal to $2j$ and their k equal to m .

The rest is the proof of Proposition 3.6. The initial page of the spectral sequence (3.6) is free, with ranks $1 + \lfloor |\sigma|/4 \rfloor$, $\lceil |\sigma|/4 \rceil$, $\lceil |\sigma|/4 \rceil$ and $\lfloor |\sigma|/4 \rfloor$ in degrees 0, 1, 2 and 3, for a total of $1 + |\sigma(K)|$. By (3.1) and Proposition 5.5, $\dim_{\mathbb{Q}} I^\natural(K; \mathbb{Q}) \geq \ell(K) = 1 + |\sigma(K)|$. Hence every differential vanishes rationally, hence integrally, and $I^\natural(K; \mathbb{Z})$ is free with these graded ranks in the grading of (3.4). Kronheimer and Mrowka's grading is this grading shifted by $\sigma(K)$ [DS24a, Thm. 8.9]. The assertions over $\mathbb{Q}$ and $\mathbb{F}_2$ follow by base change and the universal coefficient theorem. $\square$

Proof of Theorem 5.1 (iv). By (i), (ii) and Proposition 5.9, $I^\natural(K; \mathbb{Z})$ is free of rank $|\sigma(K)| + 1$, with $|\sigma(K)|/4 = 2j + M/2$. Hence $\lfloor |\sigma(K)|/4 \rfloor = 2j + M_0$ and $\lceil |\sigma(K)|/4 \rceil = 2j + M_1$, and the graded dimensions of $I^\natural(K; \mathbb{F}_2)$ are the vector of (iii). $\square$

The restriction to $n \equiv 1, 2 \pmod{6}$ comes from the pillowcase side. For $n \equiv 4, 5 \pmod{6}$ the curve of [Vue26c] contains the central circle, and the monotonicity of φ along its halves after the shear requires control of γ along them; Section 4 obtains it for $n = 5$ from the straightness of the image of the central circle (Lemma 4.7). We have not treated $n = 4$, whose twisted knots are again the pretzel knots by Lemma 5.3, nor $n \equiv 4, 5 \pmod{6}$ with $n \geq 10$.

6. DISCUSSION

6.1. Relation to earlier work. Hedden, Herald and Kirk verify their Conjecture 6.5 (Conjecture 2.5) for two-bridge knots [HHK18, §7] and for a list of torus knots [HHK18, §11]; the torus knots $T(3, n)$ are treated in [Vue26c]. For a decomposition of their own, Fukumoto, Kirk

and Pinzón-Caicedo describe the unperturbed traceless character variety of the pretzel knots and count nine generators for $P(-2, 3, 7)$ [FKP17, §5]; they do not perturb or compute a Floer complex. Theorems 1.2 and 1.3 supply the perturbed curves, their Floer complexes and the gradings. For $q \geq 9$ the differential in Theorem 1.2 is nonzero, so the agreement with $I^\natural$ is not a count of generators. The twists act on the pillowcase by the shear of [HHK18, (28)–(30)].

For Smith’s decomposition of K_q as a sum of rational tangles, Smith proves that for $q = 5$ the pairing with zero bounding cochains has rank 5 rather than $\dim I^\natural = 7$ [Smi24, proof of Thm. 5.9], so that under the conjecture of Cazassus, Herald, Kirk and Kotelskiy a nonzero bounding cochain is required there [Smi24, Thm. 5.9]. The decomposition used here needs none. Maurer–Cartan elements for Smith’s decomposition at $q = 5, 7$, and their relation to the Bar-Natan invariant of Kotelskiy, Watson and Zibrowius [KWZ19] of the tangle formed by the twist regions with 3 and q half-twists, are the subject of [Wue26d].

6.2. Open problems. *Chain level.* The pillowcase complex of Theorem 1.2 is an S-complex over $\mathbb{F}_2$ whose only nonzero component is the differential of its irreducible part (Remark 4.20). Whether it is S-chain homotopy equivalent to $\tilde{C}(K_q; \mathbb{F}_2)$ is open. For the torus knots $T(3, n)$ the corresponding statement holds [Wue26c, Thm. 1.3], and there the pillowcase differential is a component of type δ_1 ; for K_q with $q \geq 9$ it lies in the irreducible part instead.

Other families. Theorem 1.3 covers $T(3, n; 2, m)$ for $n \equiv 1, 2 \pmod{6}$. For $n \equiv 4, 5 \pmod{6}$ and $n \geq 10$ the curve of [Wue26c] has a central circle, and the shear argument of Section 4 needs that circle to be a straight line, which we prove only for $n = 5$; the signature argument also fails for $n \equiv 5 \pmod{6}$.

APPENDIX A. COMPUTATIONS

A.1. The sheared curves. Apart from the base case of Remark 4.3, a verified computation described at the end of this subsection, the proofs in Sections 4 and 5 do not use computation. The following computations corroborate them.

- (i) *Lemma 4.7.* Each displayed identity in Steps 1–7 of its proof holds identically in the indeterminates v and ζ , as verified by symbolic computation. Without the assumption $E = 0$ one has $\mathbf{Vec} \rho(w_0) \cdot \mathbf{Vec} \rho(a) = \frac{1}{2} \zeta \sin 2v E^2$ and $\mathrm{tr}(\rho(w_0) \rho(a)) = -2 \zeta \sin 2v E^2$. In the Fricke coordinates $x = \mathrm{tr} \rho(A)$, $y = \mathrm{tr} \rho(B)$, $z = \mathrm{tr} \rho(AB)$, the condition $\mathrm{tr} \rho(b) = 0$ reads $x = yz$; then $E = y^2 + z^2 - 3$,

$$\mathrm{tr} \rho(a) = -yE, \quad \mathrm{tr} \rho(w_0) = -zE, \quad \mathrm{tr} \rho(w_0 a) = yzE^2,$$

and $\mathrm{tr} \rho(w) - 2$ lies in the ideal generated by $x - yz$ and E . For the word $(ba)^{-2}c(ba)^2a$ the same reduction leaves $-y^2 - 1$, so that word is not trivial on C_0 . At three points of C_0 with algebraic coordinates, exact quaternion arithmetic gives $\rho(w) = 1$, and a numerical trace of $\Pi(C_0)$ at 50 significant digits satisfies $|\theta - 6\gamma + \pi| \leq 1.4 \cdot 10^{-47}$.

- (ii) *Combinatorics.* For every odd $q \leq 401$ we verified in exact rational arithmetic: the crossings of the sheared polygons of Corollary 4.8 with the lifts of L_0 , for both resolutions; the value of N_q ; the order of the crossing abscissae in $(\frac{\pi}{6}, \frac{5\pi}{6})$ and the absence of a crossing at $\gamma = \frac{\pi}{2}$; the bookkeeping of the graded homology in Theorem 4.18; the count of Remark 4.19; and the number of generators on the remnant circle.
- (iii) *Groups.* The identity of Lemma 4.2 holds as an identity of reduced free-group words for $n \in \{5, 8, \dots, 29\}$ and $m \leq 15$; with t_e^{+m} in place of t_e^{-m} the relator of (4.4) is never obtained. The boundary relation $ba = cd$ holds for the words (4.2), and for all $4 \leq n \leq 26$ prime to 3 and $m \leq 6$ the Alexander polynomial obtained by Fox calculus from the presentation of Lemma 4.1 with $h = -m$ equals that of the closure of $(\sigma_1 \sigma_2)^n \sigma_1^{2m}$ computed by the Burau representation. The closed form of the Alexander polynomials

in Proposition 5.4 was checked as an identity of rational functions and exactly for 1092 knots, and the determinants for $n \leq 50$ and $m \leq 20$.

- (iv) *The shear.* Applying t_e^h to the boundary values of points of the traced variety W_ε reproduces S_h to within 10^{-12} for $h = -6, \dots, 2$, whereas the shear of the opposite sign is off by about π .
- (v) *Numerical Floer complexes.* The variety W_ε is traced with the continuation code of [Wue26c]. Independent code applies S_{-m} , locates the generators on the lifts of L_0 , computes their degrees from (2.3), and enumerates the bigons whose lifts are embedded discs, as Jordan domains in $\mathbb{R}^2$. Maslov indices are computed on the unsheared curve against the line field spanned by $(1, 6 - q)$, which by Lemma 4.6(f) gives the index of the sheared curve against ℓ_1 ; an index obtained by unwrapping tangent angles along the sheared polygonal curve is unreliable near the junctions when ε is small or q is large. The results for the pretzel knots, summarized in Remark 4.21, are as follows.

- *Joined sectors.* For ten perturbations with $|5\varepsilon_A|$ between 0.002 and 0.3 and $|\varepsilon_B|$ between 0.0003 and 0.05, two earring parameters, and $q \in \{7, 9, \dots, 25, 29, 33\}$, the homology agrees with Theorem 4.18 in all 240 cases. Every bigon found runs from a down-crossing to an up-crossing on one lift of L_0 , the number of bigons is twice the rank of the differential, and the degrees agree with Proposition 4.15. The generator count $q + 2 + 4N_q$ is attained in 178 cases; in the other 62, at the larger perturbations, canceling pairs are missing, as in Remark 4.12.
- *Split sectors.* For five perturbations with $|\varepsilon_B| > \sqrt{3}|\varepsilon_A|$ and $|\varepsilon| \leq 0.031$, and $q = 7, 9, \dots, 21$, all 40 cases agree: the arc is φ -monotone with $q - 2$ generators, and the remnant circle has homology of rank 1 in each degree. The circle carries $4 + 4N_q$ generators in 37 cases; in the other three ($q = 17$, at the three larger perturbations) a canceling pair is missing. A lift of L_0 meets the lift of the circle up to seven times ($q = 21$).
- *Resolution type.* At $|\varepsilon| = 0.03$, in 24 equally spaced directions of which four lie on the excluded lines, the observed type agrees with Proposition 4.10 in all 20 remaining directions; the split ones are those at $75^\circ, 90^\circ, 105^\circ, 255^\circ, 270^\circ$ and 285° . The sample points of [Wue26c, Rem. 5.8] for $n = 5$, in both kinds of sector, also agree with Proposition 4.10.

For the twisted torus knots with $n = 7, 8, 13$ and $m = 1, 2, 3$, at two perturbations each, the graded homology agrees with Theorem 1.3 in all 18 runs.

The base case of Section 4. For Remark 4.3, the complements of $K^{(0)} \cup e$ (from a planar diagram of [HHK18, Fig. 18]) and of the closure of $(\sigma_1 \sigma_2)^5$ with an unknotted circle around two strands are compared through their canonical cell decompositions, computed with interval arithmetic by SnapPy within Sage [CDGW, HIK+16, DHL15]. The comparison lists all isometries between the two complements: there are two, and both preserve the orientation and the meridians. The surgery sign convention is checked by a Rolfsen twist: -1 surgery on the circle gives the closure of $(\sigma_1 \sigma_2)^5 \sigma_1^2$.

A.2. Reproducibility. The programs that carry out the computations of this appendix are included as ancillary files with the arXiv submission of this paper, and they are also available in the public repository

<https://github.com/bwuebben/pillowcase-bounding-cochain>.

In the ancillary files they are in the directory `code/`, and in the repository above in its directory `paper2/code/`. Each program prints the numbers of the statements it checks next to the computed values and reports whether the two agree, and the file `README.md` lists the statements that each program checks. The programs use numpy, sympy and mpmath, and the verified

computation of Remark 4.3 uses SnapPy within Sage [CDGW]. Apart from that computation, the proofs of Theorems 1.1, 1.2 and 1.3 do not depend on any of this code.

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